
UNIFYING FORMAL EXPLANATIONS: A COMPLEXITY-THEORETIC PERSPECTIVE

Anonymous authors
Anonymous submission

ABSTRACT

A broad line of work has studied the computational complexity of deriving two fundamental families of explanations for ML model predictions: (i) *sufficient reasons*, which are subsets of input features that, when fixed, determine a prediction, and (ii) *contrastive reasons*, which are subsets of input features that, when modified, alter a prediction. These explanation families have been analyzed across several modeling choices, including non-probabilistic versus probabilistic formulations and local versus global scopes. We develop a unified framework that places these settings under a common optimization view, showing that each can be characterized through minimization of a single probabilistic value-function template. We then prove that the complexity of these computations is influenced by three key properties of the value function: (i) *monotonicity*, (ii) *submodularity*, and (iii) *supermodularity* — which are three fundamental properties in *combinatorial optimization*. The resulting characterization reveals structural behavior that is not apparent when each explanation variant is considered in isolation. For instance, although the *local* value functions do not exhibit monotonicity or submodularity/supermodularity whatsoever, we demonstrate that the *global* value functions do possess these properties. This distinction supports polynomial-time guarantees for several global explanation problems across model classes that span the interpretability spectrum, including neural networks, decision trees, and tree ensembles. By contrast, the corresponding local formulations remain NP-hard even in highly simplified cases, indicating that the scope of the explanation objective itself can determine tractability.

1 INTRODUCTION

Although methods for explaining ML model decisions have advanced substantially, formal explanation guarantees across model families often come with significant computational barriers (Barceló et al., 2020; Van den Broeck et al., 2022), and these barriers are typically amplified for complex or highly non-linear model classes (Barceló et al., 2020; Adolphi et al., 2025; 2024). Consequently, the *computational complexity* of explanation generation has become a central theoretical lens for understanding how explanation guarantees scale across tasks, model classes, distributional assumptions, and local-versus-global scopes (Barceló et al., 2020; Wäldchen et al., 2021; Arenas et al., 2022; 2023; Marzouk & De La Higuera, 2024; Ordyniak et al., 2023; Laber, 2024; Bhattacharjee & Luxburg, 2024; Blanc et al., 2021; 2022).

From sufficient to contrastive reasons. Within this broader complexity-theoretic program, two explanation families have received sustained attention across model classes: (i) *sufficient reasons* and (ii) *contrastive reasons* (Barceló et al., 2020; Arenas et al., 2022; Audemard et al., 2022a; Arenas et al., 2021; Barceló et al., 2025; Marques-Silva & Ignatiev, 2022; Ignatiev et al., 2020b; Darwiche & Hirth, 2020). A *sufficient reason* is a subset of input features S whose fixed values are enough to preserve the model’s prediction under all assignments to the complementary set $\bar{S}$. A *contrastive reason* is a subset of input features S whose modification can change the model’s prediction.

Compared with additive attribution methods, which allocate importance scores across features but can be hard for humans to interpret (Kumar et al., 2020) or lack actionability (Bilodeau et al., 2024), sufficient and contrastive reasons provide *discrete, condition-like explanations* that answer questions of the form “what is enough to justify this prediction?” or “what must change to flip it?”. Because

they describe feature conditions rather than only numerical weights, they naturally support a range of explanation uses, including classic XAI methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dhurandhar et al., 2018), improved human prediction relative to additive models (Ribeiro et al., 2018; Yin & Neubig, 2022; Dhurandhar et al., 2018), and downstream tasks such as bias detection (Balkir et al., 2022; Carter et al., 2021; La Malfa et al., 2021; Muthukumar et al., 2018), model debugging (Jacovi et al., 2021), and anomaly detection (Davidson et al., 2025). Across these uses, a standard premise is that *smaller* sufficient and contrastive explanations tend to be more interpretable, making *minimality* a central guarantee of interest (Ribeiro et al., 2018; Lopardo et al., 2023; Carter et al., 2019; Barceló et al., 2020; Arenas et al., 2022; Blanc et al., 2021; Wäldchen et al., 2021).

Barceló et al. (2020) provided an early complexity analysis of deriving sufficient and contrastive reasons. That analysis established that finding the minimal-sized sufficient reason for a decision tree is NP-hard, with the complexity further increasing to Σ_2^P -hard for neural networks. In the case of contrastive reasons, they demonstrated that computing the smallest possible explanation is solvable in polynomial time for decision trees but becomes NP-Hard for neural networks. Related hardness results were later shown to extend for tree ensembles as well (Izza & Marques-Silva, 2021; Ordyniak et al., 2024; Audemard et al., 2022b).

From non-probabilistic to probabilistic explanations. A recurring criticism of the classic definitions of sufficient and contrastive reasons is that their all-or-nothing guarantees can be too rigid, as they hold in an absolute sense over entire domains and can thus lead to excessively large or uninformative explanations (Ignatiev et al., 2019; 2020a; Arenas et al., 2022; Wäldchen et al., 2021). To cover settings where absolute guarantees are unnecessarily strict, the literature has moved toward more general *probabilistic* formulations on sufficient and contrastive reasons (Wäldchen et al., 2021; Arenas et al., 2022; Blanc et al., 2021; Izza et al., 2023; Xue et al., 2023). Under this framework, the goal is to identify subsets of input features that influence a prediction with a probability exceeding a given threshold δ .

Wäldchen et al. (2021) were the first to study the complexity of probabilistic sufficient reasons, showing that for CNF classifiers the problem is NP^{PP} -hard. This hardness extends to tree ensembles and neural networks (Barceló et al., 2020; Ordyniak et al., 2023). A central theoretical insight in the probabilistic setting is the *lack of monotonicity* in the probability function (Arenas et al., 2022; Izza et al., 2023), which makes even *subset minimal* explanations computationally hard. Strikingly, (Arenas et al., 2022) show that finding a subset minimal probabilistic sufficient reason is NP-hard even for decision trees — unlike in the non-probabilistic case, where such explanations are computable in polynomial time (Huang et al., 2021; Barceló et al., 2020).

From local to global explanations. More recently, Bassan et al. (2024) extended the complexity analysis from the *local* (non-probabilistic) setting — where explanations are tied to individual predictions — to the *global* (non-probabilistic) setting, which seeks sufficient or contrastive reasons over entire domains. However, as with other non-probabilistic methods (Ignatiev et al., 2019; Barceló et al., 2020; Arenas et al., 2021; Darwiche & Hirth, 2020), the criteria are extremely strict — arguably even more so for the global formulation than in the local one. For instance, any feature excluded from a global sufficient reason is deemed strictly redundant (Bassan et al., 2024), often making the explanation span nearly all input features and thus less informative.

OUR CONTRIBUTIONS

1. We place previous explanation-computation problems — including sufficient, contrastive, probabilistic, non-probabilistic, as well as local and global — within one framework, described as a minimization task over a unified value function. We then identify three fundamental properties of the value function that significantly impact the complexity of this task: (a) *monotonicity*, (b) *supermodularity*, and (c) *submodularity*.
2. Interestingly, we establish that these properties behave in *strikingly different manners* depending on the structure of the value function. In particular, we demonstrate the surprising result that while the *local* value functions for both sufficient and contrastive reasons are *non-monotonic*, their *global* counterparts are *monotonic non-decreasing*. Moreover, we identify additional intriguing properties unique to the global setting: the global sufficient

value function is *supermodular*, whereas the global contrastive value function is *submodular* — in contrast to the local setting, where neither property holds.

3. We leverage these properties to derive new complexity results for explanation computation, revealing the intriguing finding that global explanations with guarantees can be computed efficiently, even though computing their local counterparts remains computationally hard. We demonstrate these findings across three widely used model types that span the interpretability spectrum: (i) neural networks, (ii) decision trees, and (iii) tree ensembles. First, we prove that while computing a subset-minimal local sufficient/contrastive probabilistic explanation is NP-hard even for decision trees (Arenas et al., 2022), its global counterpart can be computed in polynomial time. We further extend this result to any black-box model (including complex models such as neural networks and tree ensembles) when using empirical distributions. Specifically, we establish that obtaining a subset-minimal global sufficient/contrastive explanation is achievable in polynomial time, whereas the local version remains NP-hard for these models.
4. Finally, we present an even stronger complexity result by exploiting the submodular and supermodular properties of the value functions for the global formulation — properties that do not hold in the local case. Specifically, we establish that it is possible to achieve provable *constant-factor* approximation guarantees for computing cardinally minimal global explanations — even for complex models like neural networks or tree ensembles — when the empirical distribution is fixed. In sharp contrast, we establish strong inapproximability results for the local setting, demonstrating that no *bounded* approximation is possible, even under very simple assumptions.

Owing to space constraints, we give an overview of our main theorems and corollaries in the main text, with full proofs deferred to the appendix.

2 PRELIMINARIES

Setting. We work with an input space of dimension $n \in \mathbb{N}$, with input vectors $\mathbf{x} := (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$. Each coordinate $\mathbf{x}_i$ ranges over a feature domain $\mathcal{X}_i$, which may be discrete or continuous, so the full feature space is $\mathbb{F} := \mathcal{X}_1 \times \mathcal{X}_2 \times \dots \times \mathcal{X}_n$. The analysis is stated for classification models $f : \mathbb{F} \rightarrow [c]$, where $c \in \mathbb{N}$ is the number of classes, and for a *generic* distribution $\mathcal{D} : \mathbb{F} \rightarrow [0, 1]$ over the input space. Since accurately approximating $\mathcal{D}$ may itself be computationally infeasible in many application settings, the framework also covers the common empirical alternative of working with a fixed dataset $\mathbf{D} := \{\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}\}$ as a proxy for the underlying distribution, for instance when $\mathbf{D}$ is sampled from available training data.

The framework covers both local and global explanation scopes. In the local case, an explanation targets a specific prediction $\mathbf{x} \in \mathbb{F}$, providing a form of reasoning for why the model predicted $f(\mathbf{x})$ for that instance. In the global case, explanations aim to reflect the general reasoning behind the behavior of f across a wider region of the input space, independent of any specific $\mathbf{x}$, and to characterize its overall decision-making logic.

Models. Several results in the paper are model-agnostic, including the structural properties of the value functions, while the complexity consequences are instantiated for representative and widely used model families. To cover both interpretable and black-box regimes without changing the formal task, we analyze: (i) decision trees, (ii) neural networks (all architectures at least as expressive as feed-forward ReLU networks), and (iii) tree ensembles, including majority-voting ensembles (e.g., random forests) and weighted-voting ensembles (e.g., XGBoost). Formal definitions are provided in the Appendix.

Distributions. In probabilistic explanation settings, the computational picture depends not only on the model class but also on how the input distribution $\mathcal{D}$ is represented. We therefore distinguish three distribution regimes: (i) *general distributions*, which make no specific assumptions over $\mathcal{D}$ and thus encompass all possible distributions. We use these distributions mainly in proofs of properties that hold universally; (ii) *empirical distributions*, which include all distributions derived from the finite dataset $\mathbf{D}$ — an approach commonly employed in XAI (Lundberg & Lee, 2017; Van den Broeck et al., 2022); and (iii) *independent distributions*, which assume that features in $\mathcal{D}$ are mutually independent — another widely adopted assumption in XAI literature (Arenas et al., 2022; 2023;

Lundberg & Lee, 2017; Ribeiro et al., 2018). We note that empirical distributions do not necessarily imply feature independence; rather, they can represent complex dependencies extracted from finite datasets (Van den Broeck et al., 2022). The complete formal definitions of these distributions are provided in the Appendix.

3 FORMS OF REASONS

This section formalizes the explanation types covered by our framework, starting with the strict non-probabilistic definitions of (local/global) sufficient and contrastive reasons, and then extending to their more flexible and generalizable probabilistic counterparts.

3.1 Non-Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

Sufficient reasons. In feature-selection views of explanation, one often seeks a small set of features that carries enough information to justify a model decision. We study the established *sufficiency* criterion for this purpose, which underlies several explainability methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dasgupta et al., 2022). The same criterion can be applied at different scopes: locally for one prediction, or globally over the input domain. Following standard conventions, a *local sufficient reason* is a subset of features $S \subseteq \{1, \dots, n\}$ such that fixing the features in S to their values in $\mathbf{x}$ preserves the prediction $f(\mathbf{x})$ for every assignment of the remaining features $\bar{S}$. Formally, S is a local sufficient reason for $\langle f, \mathbf{x} \rangle$ iff the following condition holds:

$$\forall \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (1)$$

Here, $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ denotes a vector whose coordinates in S come from $\mathbf{x}$ and whose remaining coordinates come from $\mathbf{z}$. This definition also connects the formal reason notion to *semi-factual* explanations (Kenny & Keane, 2021; Alfano et al., 2025), which search for nearby alternatives $\mathbf{z}'$ that keep the prediction unchanged, i.e., $f(\mathbf{z}') = f(\mathbf{x})$. When $\mathbf{z}'$ is instantiated as $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ and distance is measured by Hamming distance, the two notions coincide.

The corresponding global notion asks for a feature subset whose sufficiency is not tied to a single instance. A *global sufficient reason* (Bassan et al., 2024) is a subset $S \subseteq \{1, \dots, n\}$ that serves as a local sufficient reason for *every* input $\mathbf{x} \in \mathbb{F}$:

$$\forall \mathbf{x}, \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (2)$$

Contrastive reasons. A complementary explanation style identifies which features can *modify* a prediction (Dhurandhar et al., 2018; Mothilal et al., 2020; Guidotti, 2024). Rather than asking what is enough to preserve the current output, this criterion asks which feature changes can alter it. Formally, a subset $S \subseteq \{1, \dots, n\}$ is a *local contrastive reason* for $\langle f, \mathbf{x} \rangle$ iff:

$$\exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (3)$$

Contrastive reasons are also closely connected to *counterfactual* explanations (Guidotti, 2024; Mothilal et al., 2020), which seek a nearby assignment $\mathbf{x}'$ for which the prediction flips, i.e., $f(\mathbf{x}') \neq f(\mathbf{x})$. When the distance metric is taken to be the Hamming weight, the two notions coincide.

As with sufficiency, contrastive reasoning can be lifted from a single prediction to a domain-level requirement. The resulting global contrastive reason identifies features whose alteration can change predictions throughout the domain of interest, and is closely related to bias-identification methods (Arenas et al., 2021; Bassan et al., 2024; Darwiche & Hirth, 2020) and *group* counterfactual explanations over multiple data instances (Carrizosa et al., 2024; Warren et al., 2024). Formally, S is a *global contrastive reason* with respect to f iff:

$$\forall \mathbf{x} \in \mathbb{F}, \quad \exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (4)$$

3.2 Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

As discussed in the introduction, non-probabilistic sufficient and contrastive reasons are often criticized for imposing significantly overly strict conditions, motivating a shift to *probabilistic* definitions (Arenas et al., 2022; Ribeiro et al., 2018; Izza et al., 2023; Wäldchen et al., 2021; Bounia & Koriche, 2023; Subercaseaux et al., 2025; Wang et al., 2021a; Blanc et al., 2021), which generalize these requirements by demanding that the guarantees hold with probability at least $\delta \in [0, 1]$. The special case $\delta = 1$ recovers the original non-probabilistic definitions.

Probabilistic sufficient reasons. We define $S \subseteq \{1, \dots, n\}$ as a *local δ -sufficient reason* with respect to $\langle f, \mathbf{x} \rangle$ if, when the features in S are fixed to their corresponding values in $\mathbf{x}$ and the remaining features are sampled from a distribution $\mathcal{D}$, the classification remains unchanged with probability at least δ . In other words:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) \geq \delta. \quad (5)$$

where $\mathbf{z}_S = \mathbf{x}_S$ denotes that the features in S of vector $\mathbf{z}$ are fixed to their corresponding values in $\mathbf{x}$. We adopt the standard notion of global explanations — by averaging over all inputs in the global domain — and define a subset $S \subseteq \{1, \dots, n\}$ as a *global δ -sufficient reason* with respect to $\langle f \rangle$ if, when taking the expectation of the local sufficiency probability over samples drawn from the distribution $\mathcal{D}$, the expectation remains with value at least δ . In other words:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \geq \delta. \quad (6)$$

Probabilistic contrastive reasons. In parallel with the non-probabilistic case, we define a *local δ -contrastive reason* as a subset of input features that changes a prediction with some probability. Here, unlike sufficient reasons, we set the features of the *complementary* set $\bar{S}$ to their respective values in $\mathbf{x}$, and when allowing the features in S to vary, we require the prediction to differ from the original prediction $f(\mathbf{x})$ with a probability exceeding δ . Formally:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}) \geq \delta. \quad (7)$$

For the global setting, we define a subset S to be a *global δ -contrastive reason*, analogous to global sufficient reasons, by computing the expectation over all local contrastive reasons sampled from the distribution $\mathcal{D}$, and requiring that this expected value exceeds δ :

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (8)$$

4 FORMS OF MINIMALITY

Across the explanation forms above — sufficient or contrastive, local or global, probabilistic or non-probabilistic — the literature commonly treats smaller feature subsets as more interpretable. Minimality is therefore a cross-cutting guarantee rather than a property of one particular explanation definition. We study two standard minimality notions that apply throughout the framework:

Definition 1. Assuming a subset $S \subseteq \{1, \dots, n\}$ is an explanation, then:

1. S is a *cardinally-minimal explanation* (Barceló et al., 2020; Bassan et al., 2024) iff S has the smallest explanation cardinality $|S|$ (i.e., there is no explanation S' such that $|S'| < |S|$).
2. $S \subseteq \{1, \dots, n\}$ is a *subset-minimal explanation* (Arenas et al., 2022; Ignatiev et al., 2019) iff S is an explanation, and any $S' \subsetneq S$ is not an explanation.

Cardinal minimality is strictly stronger than subset minimality: every cardinally minimal S is subset minimal, but not vice versa (see Appendix B.3 for an example). We use the terms subset minimal and cardinally minimal, rather than local and global minima, to avoid confusion with the separate local/global distinction for explanations. Both minimality notions apply to *all* explanation forms in the framework. For example, a cardinally minimal local probabilistic δ -sufficient reason has the smallest possible $|S|$, whereas a subset-minimal one is any qualifying S for which no proper subset $S' \subsetneq S$ also qualifies.

5 A UNIFIED COMBINATORIAL OPTIMIZATION TASK

The explanation variants introduced above — local or global, sufficient or contrastive, probabilistic or non-probabilistic — share the same combinatorial core: find a minimal-size subset S whose value reaches a threshold, $v(S) \geq \delta$, with the non-probabilistic variants recovered by setting $\delta := 1$. We use v_{suff}^{ℓ} for the local sufficiency probability in equation 5, i.e., $\Pr_{\mathbf{z} \sim \mathcal{D}_P}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)$, and v_{suff}^g for its global analogue in equation 6. Likewise, v_{con}^{ℓ} and v_{con}^g denote the local and global contrastive probabilities from equations 7 and 73. This notation lets us state a single optimization task that covers the cardinally minimal versions of all four explanation forms:

Cardinally Minimal δ -Explanation:

Input: Model f , a distribution $\mathcal{D}$, (possibly, an input $\mathbf{x}$), a *general* value function $v : 2^n \rightarrow [0, 1]$ (defined using f , $\mathcal{D}$, and possibly $\mathbf{x}$), and some $\delta \in [0, 1]$.

Output: A subset $S \subseteq [n]$ such that $v(S) \geq \delta$ and $|S|$ is minimal.

Similarly, for the relaxed condition where the goal is to obtain a subset-minimal rather than a cardinally-minimal local or global sufficient/contrastive reason, we define the following relaxed optimization objective:

Subset Minimal δ -Explanation:

Input: Model f , a distribution $\mathcal{D}$, (possibly, an input $\mathbf{x}$), a *general* value function $v : 2^n \rightarrow [0, 1]$ (defined using f , $\mathcal{D}$, and possibly $\mathbf{x}$), and some $\delta \in [0, 1]$.

Output: A subset $S \subseteq [n]$ such that $v(S) \geq \delta$ and for any $S' \subseteq S$ it holds that $v(S') < \delta$.

Properties of v that affect the complexity. The unified task makes the role of value-function structure explicit. We focus on properties that are central in combinatorial optimization and later determine which explanation scopes admit efficient algorithms or approximation guarantees. The first is *non-decreasing monotonicity*, which ensures that the marginal contribution $v(S \cup \{i\}) - v(S)$ is consistently non-negative. Formally:

Definition 2. We say that a value function v maintains non-decreasing monotonicity if for any $S \subseteq \{1, \dots, n\}$ and any $i \in \{1, \dots, n\}$ it holds that: $v(S \cup \{i\}) \geq v(S)$.

The other key properties of *supermodularity* and *submodularity* pertain to the behavior of the marginal contribution $v(S \cup \{i\}) - v(S)$. Specifically, in the supermodular case, this contribution forms a monotone non-decreasing function. In contrast, under the dual definition of the *submodular* case, the marginal contribution $v(S \cup \{i\}) - v(S)$ is a monotone non-increasing function. Formally:

Definition 3. Let there be some value function v , some $S \subseteq S' \subseteq \{1, \dots, n\}$, and $i \notin S'$. Then:

1. v maintains supermodularity iff it holds that: $v(S \cup \{i\}) - v(S) \leq v(S' \cup \{i\}) - v(S')$.
2. v maintains submodularity iff it holds that: $v(S \cup \{i\}) - v(S) \geq v(S' \cup \{i\}) - v(S')$.

6 UNRAVELING NEW PROPERTIES OF THE GLOBAL VALUE FUNCTIONS

For local non-probabilistic explanations, prior work shows that subset-minimal sufficient or contrastive reasons can be computed because monotonicity holds in the restricted threshold case $\delta = 1$. That setting is useful as a baseline but narrow in scope: it requires an absolute guarantee and does not cover probabilistic tolerances. When the same local objectives are generalized to arbitrary δ , prior results show that monotonicity breaks down, making explanation computation substantially harder (Arenas et al., 2022; Kozachinskiy, 2023; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b).

At the even more extreme case of the *global* and *non-probabilistic* setting, Bassan et al. (2024) demonstrated the stringent *uniqueness* property — i.e., there is exactly *one* subset-minimal explanation. However, requiring $\delta = 1$ makes this setting highly restrictive, especially in the global case, where the explanation conditions must hold for *all* inputs. In fact, Bassan et al. (2024) proves that this unique minimal subset is equivalent to the subset of all features that are not strictly redundant (which may, in practice, be all of them). We prove that in the general global *probabilistic* setting — for any δ — this uniqueness property actually does *not* hold, and the number of subsets can even be *exponential*.

Proposition 1. *While the non-probabilistic case ($\delta = 1$) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary δ), there exist functions f that have $\Theta(\frac{2^n}{\sqrt{n}})$ subset-minimal global sufficient reasons.*

Interestingly, although the uniqueness property fails to hold in the general case for arbitrary δ (and is restricted to the special case of $\delta = 1$), we establish that the crucial *monotonicity* property holds for the global value function across all values of δ . This applies to both the sufficient (v_{suff}^g) and contrastive (v_{con}^g) value functions. This finding is surprising as it stands in sharp contrast to the local setting, where the corresponding value functions (v_{suff}^ℓ and v_{con}^ℓ) do not satisfy this property:

Proposition 2. *While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.*

Beyond the surprising insight that monotonicity holds for the global value functions — but fails to hold in the local one — we identify additional structural properties unique to the global setting, including submodularity or supermodularity. In particular, we establish that under the common assumption of feature independence, the global sufficient value function v_{suff}^g exhibits supermodularity. In contrast, its local counterpart v_{suff}^ℓ fails to exhibit this property even under the much more restrictive assumption of a uniform (and hence independent) input distribution. Specifically:

Proposition 3. *While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{suff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{suff}^g is supermodular.*

Interestingly, for the second family of explanation settings — specifically, that of obtaining a global probabilistic *contrastive* reason — we establish that the corresponding value function is not supermodular, but rather *submodular*. This result is particularly surprising when contrasted with the local setting, where the value function is neither submodular nor supermodular.

Proposition 4. *While the local probabilistic contrastive setting (for any δ) lacks submodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{con}^ℓ is not submodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{con}^g is submodular.*

7 COMPUTATIONAL COMPLEXITY RESULTS

7.1 SUBSET MINIMAL EXPLANATIONS

This section examines the complexity of obtaining *subset minimal* explanations across local/global and sufficient/contrastive variants for the model types analyzed. The key structural property is *monotonicity* of the value function: the previous section established monotonicity for both global value functions, v_{con}^g and v_{suff}^g , but not for the local value functions, v_{con}^ℓ and v_{suff}^ℓ . This distinction explains why a greedy algorithm converges to a subset-minimal explanation for the global formulation but not for the local formulation. The resulting complexity picture is broad across the studied explanation forms: local subset-minimal explanations remain hard, whereas global subset-minimal explanations are tractable. We begin with the following generalized greedy algorithm:

Algorithm 1 Subset Minimal Explanation Search

Input Value function v , and some $\delta \in [0, 1]$

```

1:  $S \leftarrow \{1, \dots, n\}$ 
2: while  $\min_{i \in S} v(S \setminus \{i\}) \geq \delta$  do
3:    $j \leftarrow \arg\max_{i \in S} v(S \setminus \{i\})$ 
4:    $S \leftarrow S \setminus \{j\}$ 
5: end while
6: return  $S$ 

```

$\triangleright S$ is a (subset minimal?) δ -explanation

Algorithm 1 aims to obtain a subset-minimal δ -explanation. We begin the algorithm with the subset S initialized as the entire input space $\{1, \dots, n\}$. Iteratively, we check whether the minimal value that

$v(S \setminus \{i\})$ can attain exceeds δ . In each iteration, we remove a feature j from S that minimizes the decrease in the value function, selecting the feature j that maximizes $v(S \setminus \{j\})$. Once this iterative process concludes, we return S .

The key determinant of whether Algorithm 1 yields a subset-minimal explanation is the *monotonicity* property of the value function v . The algorithm concludes with a phase in which removing any individual feature from S results in $v(S \setminus \{i\})$ being smaller than δ . However, monotonicity ensures that this holds for any $v(S \setminus S')$, providing a significantly stronger guarantee. Given the monotonicity property of the value functions established in the previous sections, we derive the following proposition:

Proposition 5. *Computing Algorithm 1 with the local value functions v_{con}^l and v_{suff}^l does not always converge to a subset minimal δ -sufficient/contrastive reason. However, computing it with the global value functions v_{con}^g or v_{suff}^g necessarily produces subset minimal δ -sufficient/contrastive reasons.*

Building on this result, we proceed to establish new complexity findings for deriving various forms of subset-minimal explanations within our framework, considering the different analyzed model types.

Decision trees. We first instantiate the framework in an interpretable model class, assuming that f is a decision tree and that $\mathcal{D}$ is independent. Even in this relatively structured regime, prior work shows a local hardness barrier: unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm computes a subset-minimal local δ -sufficient reason for decision trees under independent distributions, including the uniform case (Arenas et al., 2022). The global formulation behaves differently. Under the same model family and compatible distributional assumptions, a subset-minimal global δ -sufficient reason for decision trees can be computed in polynomial time. Formally:

Theorem 1. *If f is a decision tree and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time. However, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm exists for computing a local δ -sufficient reason for decision trees even under independent distributions.*

Extension to other tractable models. We additionally note that a similar tractability guarantee for computing global explanations also holds for *orthogonal DNFs* (Crama & Hammer, 2011), which we briefly recall are Disjunctive Normal Form formulas whose terms are pairwise mutually exclusive. Establishing the result for this class is useful more generally because orthogonal DNFs *generalize* decision trees while preserving their clean structural properties, allowing our guarantees to extend beyond tree-structured models. For completeness, we give the full argument in Appendix N.

Neural networks, tree ensembles, and other complex models. The same global/local separation is not limited to decision trees. For empirical distributions, the global subset-minimal tasks depend only on evaluating the model over a finite dataset; thus, under the standard assumption that f has polynomial-time inference, subset-minimal global δ -sufficient and δ -contrastive reasons can be computed in polynomial time.

Proposition 6. *For any model f , and empirical distribution $\mathcal{D}$ — computing a subset-minimal global δ -sufficient or δ -contrastive reason for f can be done in polynomial time.*

Because this tractability statement is model-agnostic under empirical distributions, it yields a broad contrast with local explanation computation. For expressive model classes, subset-minimal *local* explanations remain intractable even under empirical distributions. We establish this separation for neural networks and tree ensembles:

Theorem 2. *Assuming f is a neural network or a tree ensemble, and $\mathcal{D}$ is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global δ -sufficient and contrastive reasons. However, unless $\text{PTIME}=\text{NP}$, there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason or a subset minimal local δ -contrastive reason.*

7.2 APPROXIMATE CARDINALLY MINIMAL EXPLANATIONS

This subsection moves from subset minimality to the stronger requirement of finding a *cardinally minimal* δ -sufficient or δ -contrastive reason. Under the global formulation, the combination of monotonicity with submodularity or supermodularity brings these explanation tasks into contact with

well-studied covering problems, yielding *provable approximation* guarantees. The corresponding local formulations do not inherit this structure and remain intractable, reinforcing that the explanation scope is a decisive part of the computational problem.

A unified greedy approximation algorithm. With a non-decreasing monotone submodular value function, the cardinal-minimal explanation problem aligns with *submodular set cover* (Wolsey, 1982); with a supermodular value function, it aligns with a related non-submodular covering variant (Shi et al., 2021). These covering problems are useful here because they provide approximation guarantees that can be transferred to global explanation computation (Wolsey, 1982; Iyer & Bilmes, 2013; Chen & Crawford, 2023). We therefore use the greedy procedure in Algorithm 2 as a common approximation template for cardinally minimal sufficient and contrastive δ -reasons, with guarantees depending on which value-function structure applies.

Algorithm 2 Cardinally Minimal Explanation Approximation Search

Input Value function v , and some $\delta \in [0, 1]$

```

1:  $S \leftarrow \emptyset$ 
2: while  $\max_{i \in S} v(S \cup \{i\}) < \delta$  do
3:    $j \leftarrow \operatorname{argmax}_{i \in S} v(S \cup \{i\})$ 
4:    $S \leftarrow S \cup \{j\}$ 
5: end while
6: return  $S$   $\triangleright |S|$  is a (provable approximation?) of a cardinally minimal  $\delta$ -explanation

```

Algorithm 2 closely resembles Algorithm 1, but works bottom-up rather than top-down. It starts with an empty subset and incrementally adds features, each time selecting the one that minimizes the increase in $v(S \cup i)$, stopping when adding any feature would push the value over δ .

Cardinally minimal contrastive reasons. We begin with global contrastive reasons, where monotonicity and submodularity hold, reducing the task to the classic *submodular set cover* problem and allowing us to apply Wolsey (1982)’s classic guarantee via Algorithm 2. For integer-valued objectives, the algorithm achieves a Harmonic-based factor, and more generally an $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation. Under an *empirical* distribution $\mathcal{D}$ with fixed sample size, this becomes a *constant approximation*, only *logarithmic* in the sample size, yielding a substantially strong guarantee. By contrast, in the *local* setting — even for a *single* sample point — no *bounded* approximation exists, marking a sharp gap between global and local cases.

Theorem 3. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, no bounded approximation exists for computing a local cardinally minimal δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

We also provide a matching *lower bound* for the bound in Theorem 3 in the specific case of an empirical distribution without any independence assumption (see Appendix M).

Cardinally minimal sufficient reasons. Unlike the submodular set cover problem — linked to cardinally minimal global *contrastive* reasons and admitting strong approximations — the supermodular variant, tied to global *sufficient* reasons, is harder to approximate. Still, it offers guarantees when the function has bounded curvature (Shi et al., 2021). The *total curvature* of a function $v : 2^n \rightarrow \mathbb{R}$ is defined as $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$. Leveraging results from Shi et al. (2021), we establish that Algorithm 2 achieves an $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation.

Notably, under a fixed empirical distribution, the approximation becomes constant. While contrastive reasons admit a tighter $\mathcal{O}(\ln(|D|))$ bound, sufficient reasons incur an extra $\frac{1}{1-k^f}$ factor — yet still yield a constant-factor approximation. In sharp contrast, the local variant remains inapproximable, lacking any *bounded* approximation even when $|D| = 1$.

Theorem 4. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$ -approximation for*

computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence. In contrast, unless $PTIME=NP$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.

Taken together, these findings strengthen Subsection 7.1, which showed the tractability of computing subset-minimal global explanations in stark contrast to local ones. Here, we further show that approximating cardinally minimal global explanations is tractable, unlike their inapproximable local counterparts.

8 LIMITATIONS AND FUTURE WORK

Many of the central findings — especially the value-function properties — are stated at a general level, while the algorithmic and hardness consequences are instantiated for particular model classes (e.g., neural networks), distributional assumptions (e.g., empirical distributions), and explanation definitions covered by the framework. This design clarifies which conclusions are structural and which depend on the chosen model or distribution representation. Other combinations of models, distributions, and explanation objectives remain open, but the results give a foundation for extending the same analysis to broader explainability scenarios.

More concretely, our study brings forward several important open theoretical questions:

1. Firstly, it would be interesting to investigate whether some of the tractable complexity results we obtained for global explainability can be tightened for simpler model classes, such as decision trees, linear models, or other models with inherently tractable structure. Although local explanations for such models are often intractable to compute (Arenas et al., 2022; Subercaseaux et al., 2025), the global explanation forms we study in this work exhibit strong structural properties, enabling significantly improved complexity guarantees that may be tightened even further. Furthermore, tightening the approximation guarantees when working with empirical distributions represents a key open direction.
2. Secondly, when *exact* computation of expectations is infeasible, it is natural to ask whether alternative techniques — such as Fully Polynomial Randomized Approximation Schemes (FPRAS) for model classes like DNF formulas (Meel et al., 2019) (and thus for tree ensembles), or Monte Carlo-based approximations as the ones used in (Subercaseaux et al., 2025) — can be employed to approximate these expectations. A key open challenge is determining how to maintain minimality guarantees under approximation error, and what tolerance levels still allow the guarantees to hold.
3. Finally, we believe that studying alternative notions of feature importance — e.g., other loss measures such as KL divergence (Conmy et al., 2023) — may reveal similar patterns of monotonicity, submodularity, or supermodularity when transitioning from local to global explanations. Such properties could enable efficient feature selection under these alternative metrics as well.

9 CONCLUSION

We present a unified framework that treats sufficient and contrastive reasons, probabilistic and non-probabilistic guarantees, and local and global scopes as instances of a common value-function optimization problem. Within this framework, the local variants lack the structural properties most useful for combinatorial optimization — monotonicity, submodularity, and supermodularity — whereas the global variants exhibit the relevant structure: (i) *monotonicity*, (ii) *submodularity* in the case of contrastive reasons, and (iii) *supermodularity* for sufficient reasons. These properties support tractability and approximation results for global explanations across the model classes and distributions studied here, including complex model classes such as neural networks. The corresponding local formulations remain NP-hard, even in simplified scenarios. Taken together, the results identify scope as a central determinant of explanation complexity and provide a basis for analyzing additional formal explanation settings.

REFERENCES

- Federico Adolphi, Martina G Vilas, and Todd Wareham. Complexity-Theoretic Limits on the Promises of Artificial Neural Network Reverse-Engineering. In *Proc. 46th Annual Meeting of the Cognitive Science Society (CogSci)*, 2024.
- Federico Adolphi, Martina Vilas, and Todd Wareham. The Computational Complexity of Circuit Discovery for Inner Interpretability. In *Proc. 13th Int. Conf. on Learning Representations (ICLR)*, 2025.
- Gianvincenzo Alfano, Sergio Greco, Domenico Mandaglio, Francesco Parisi, Reza Shahbazian, and Irina Trubitsyna. Even-If Explanations: Formal Foundations, Priorities and Complexity. In *Proc. 39th AAAI Conf. on Artificial Intelligence*, pp. 15347–15355, 2025.
- Ibrahim M Almanjahie, Zouaoui Chikr Elmezouar, Ali Laksaci, and Mustapha Rachdi. KNN Local Linear Estimation of the Conditional Cumulative Distribution Function: Dependent Functional Data Case. *Comptes Rendus. Mathématique*, 356(10):1036–1039, 2018.
- David Alvarez Melis and Tommi Jaakkola. Towards Robust Interpretability with Self-Explaining Neural Networks. In *Proc. 32nd Conf. on Neural Information Processing Systems (NeurIPS)*, 2018.
- Leila Amgoud and Martin Cooper. Axiomatic Foundations of Counterfactual Explanations. 2026. Technical Report. <https://arxiv.org/abs/2602.04028>.
- Guy Amir, Shahaf Bassan, and Guy Katz. Hard to Explain: On the Computational Hardness of In-Distribution Model Interpretation. In *Proc. 27th European Conf. on Artificial Intelligence (ECAI)*, 2024.
- Alaa Anani, Tobias Lorenz, Mario Fritz, and Bernt Schiele. Pixel-level Certified Explanations via Randomized Smoothing. In *Proc. 42nd Int. Conf. on Machine Learning (ICML)*, 2025.
- Marcelo Arenas, D. Baez, P. Barceló, J. Pérez, and B. Subercaseaux. Foundations of Symbolic Languages for Model Interpretability. In *Proc. 34th Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp. 11690–11701, 2021.
- Marcelo Arenas, P. Barceló, M. Romero, and B. Subercaseaux. On Computing Probabilistic Explanations for Decision Trees. In *Proc. 35th Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp. 28695–28707, 2022.
- Marcelo Arenas, Pablo Barceló, Leopoldo Bertossi, and Mikaël Monet. On the Complexity of SHAP-Score-Based Explanations: Tractability via Knowledge Compilation and Non-Approximability Results. *Journal of Machine Learning Research (JMLR)*, 24(63):1–58, 2023.
- Gilles Audemard, S. Bellart, L. Bounia, F. Koriche, J. Lagniez, and P. Marquis. On the Computational Intelligibility of Boolean Classifiers. In *Proc. 18th Int. Conf. on Principles of Knowledge Representation and Reasoning (KR)*, pp. 74–86, 2021.
- Gilles Audemard, Steve Bellart, Louenas Bounia, Frédéric Koriche, Jean-Marie Lagniez, and Pierre Marquis. On Preferred Abductive Explanations for Decision Trees and Random Forests. In *Proc. 31st Int. Joint Conf. on Artificial Intelligence (IJCAI)*, pp. 643–650, 2022a.
- Gilles Audemard, Steve Bellart, Louenas Bounia, Frédéric Koriche, Jean-Marie Lagniez, and Pierre Marquis. Trading Complexity for Sparsity in Random Forest Explanations. In *Proc. 36th AAAI Conf. on Artificial Intelligence*, number 5, pp. 5461–5469, 2022b.
- Gilles Audemard, Jean-Marie Lagniez, Pierre Marquis, and Nicolas Szczepanski. Computing Abductive Explanations for Boosted Trees. In *Proc. 26th Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*, pp. 4699–4711, 2023.
- Steve Azzolin, Sagar Malhorta, Andrea Passerini, and Stefano Teso. Beyond Topological Self-Explainable GNNs: A Formal Explainability Perspective. In *Proc. 42nd Int. Conf. on Machine Learning (ICML)*, 2025.

-
- Esma Balkir, Isar Nejadgholi, Kathleen C Fraser, and Svetlana Kiritchenko. Necessity and Sufficiency for Explaining Text Classifiers: A Case Study in Hate Speech Detection. In *Proc. Conf. of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pp. 2672–2686, 2022.
- Pablo Barceló, M. Monet, J. Pérez, and B. Subercaseaux. Model Interpretability through the Lens of Computational Complexity. In *Proc. 33rd Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp. 15487–15498, 2020.
- Pablo Barceló, Alexander Kozachinskiy, Miguel Romero, Bernardo Subercaseaux, and José Verschae. Explaining k-Nearest Neighbors: Abductive and Counterfactual Explanations. In *Proc. 3rd ACM on Management of Data*, pp. 1–26, 2025.
- Shahaf Bassan and G. Katz. Towards Formal XAI: Formally Approximate Minimal Explanations of Neural Networks. In *Proc. 29th Int. Conf. on Tools and Algorithms for the Construction and Analysis of Systems (TACAS)*, pp. 187–207, 2023.
- Shahaf Bassan, Guy Amir, Davide Corsi, Idan Refaeli, and Guy Katz. Formally Explaining Neural Networks Within Reactive Systems. In *Proc. 23rd Int. Conf. on Formal Methods in Computer-Aided Design (FMCAD)*, pp. 1–13, 2023.
- Shahaf Bassan, Guy Amir, and Guy Katz. Local vs. Global Interpretability: A Computational Complexity Perspective. In *Proc. 41st Int. Conf. on Machine Learning (ICML)*, 2024.
- Shahaf Bassan, Guy Amir, Meirav Zehavi, and Guy Katz. What makes an Ensemble (Un) Interpretable? In *Proc. 42nd Int. Conf. on Machine Learning (ICML)*, 2025a.
- Shahaf Bassan, Yizhak Yisrael Elboher, Tobias Ladner, Matthias Althoff, and Guy Katz. Explaining, Fast and Slow: Abstraction and Refinement of Provable Explanations. In *Proc. 42nd Int. Conf. on Machine Learning (ICML)*, 2025b.
- Shahaf Bassan, Ron Eliav, and Shlomit Gur. Explain Yourself, Briefly! Self-Explaining Neural Networks with Concise Sufficient Reasons. In *Proc. 13th Int. Conf. on Learning Representations (ICLR)*, 2025c.
- Shahaf Bassan, Shlomit Gur, Sergey Zeltyn, Konstantinos Mavrogiorgos, Ron Eliav, and Dimosthenis Kyriazis. Self-Explaining Neural Networks for Business Process Monitoring. 2025d. Technical Report. <https://arxiv.org/abs/2503.18067>.
- Shahaf Bassan, Michal Moshkovitz, and Guy Katz. Additive Models Explained: A Computational Complexity Approach. In *Proc. 39th Conf. on Neural Information Processing Systems (NeurIPS)*, 2025e.
- Shahaf Bassan, Yizhak Yisrael Elboher, Tobias Ladner, Volkan Şahin, Jan Kretinsky, Matthias Althoff, and Guy Katz. Provably Explaining Neural Additive Models. In *Proc. 14th Int. Conf. on Learning Representations (ICLR)*, 2026.
- Robi Bhattacharjee and Ulrike Luxburg. Auditing Local Explanations is Hard. In *Proc. 38th Conf. on Neural Information Processing Systems (NeurIPS)*, pp. 18593–18632, 2024.
- Blair Bilodeau, Natasha Jaques, Pang Wei Koh, and Been Kim. Impossibility Theorems for Feature Attribution. *Proc. of the National Academy of Sciences (PNAS)*, 121(2):e2304406120, 2024.
- Guy Blanc, Jane Lange, and Li-Yang Tan. Provably Efficient, Succinct, and Precise Explanations. In *Proc. 35th Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp. 6129–6141, 2021.
- Guy Blanc, Caleb Koch, Jane Lange, and Li-Yang Tan. A Query-Optimal Algorithm for Finding Counterfactuals. In *Proc. 39th Int. Conf. on Machine Learning (ICML)*, pp. 2075–2090, 2022.
- Ryma Boumazouza, Fahima Cheikh-Alili, Bertrand Mazure, and Karim Tabia. ASTERYX: A model-Agnostic SaT-basEd appRoach for SYmbolic and Score-Based EXplanations. In *Proc. 30th ACM Int. Conf. on Information & Knowledge Management (CIKM)*, pp. 120–129, 2021.

-
- Ryma Boumazouza, Raya Elsaleh, Melanie Ducoffe, Shahaf Bassan, and Guy Katz. FAME: Formal Abstract Minimal Explanation for Neural Networks. In *Proc. 14th Int. Conf. on Learning Representations (ICLR)*, 2026.
- Louenas Bounia. Enhancing the Intelligibility of Boolean Decision Trees with Concise and Reliable Probabilistic Explanations. In *Proc. 20th Int. Conf. on Information Processing and Management of Uncertainty in Knowledge-Based Systems (IPMU)*, pp. 205–218, 2024.
- Louenas Bounia. Using Submodular Optimization to Approximate Minimum-Size Abductive Path Explanations for Tree-Based Models. In *Proc. 41st Conf. on Uncertainty in Artificial Intelligence (UAI)*, 2025.
- Louenas Bounia and Frederic Koriche. Approximating Probabilistic Explanations via Supermodular Minimization. In *Proc. 39th Conf. on Uncertainty in Artificial Intelligence (UAI)*, pp. 216–225, 2023.
- Marco Calautti, Enrico Malizia, and Cristian Molinaro. On the Complexity of Global Necessary Reasons to Explain Classification. 2025. Technical Report. <https://arXiv.org/abs/2501.06766>.
- Emilio Carrizosa, Jasone Ramírez-Ayerbe, and Dolores Romero Morales. Mathematical Optimization Modelling for Group Counterfactual Explanations. *European Journal of Operational Research (EJOR)*, 319(2):399–412, 2024.
- Brandon Carter, Jonas Mueller, Siddhartha Jain, and David Gifford. What Made You Do This? Understanding Black-Box Decisions with Sufficient Input Subsets. In *Proc. 22nd Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*, pp. 567–576, 2019.
- Brandon Carter, Siddhartha Jain, Jonas W Mueller, and David Gifford. Overinterpretation Reveals Image Classification Model Pathologies. In *Proc. 35th Conf. on Neural Information Processing Systems (NeurIPS)*, pp. 15395–15407, 2021.
- Wenjing Chen and Victoria Crawford. Bicriteria Approximation Algorithms for the Submodular Cover Problem. In *Proc. 37th Conf. on Neural Information Processing Systems (NeurIPS)*, pp. 72705–72716, 2023.
- Arthur Conmy, Augustine Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. Towards Automated Circuit Discovery for Mechanistic Interpretability. *Proc. 37th Conf. on Neural Information Processing Systems (NeurIPS)*, pp. 16318–16352, 2023.
- Yves Crama and Peter L Hammer. *Boolean Functions: Theory, Algorithms, and Applications*. Cambridge University Press, 2011.
- Adnan Darwiche. Logic for Explainable AI. In *Proc. 38th ACM/IEEE Symposium on Logic in Computer Science (LICS)*, pp. 1–11, 2023.
- Adnan Darwiche and Auguste Hirth. On the Reasons Behind Decisions. In *Proc. 24th European Conf. on Artificial Intelligence (ECAI)*, pp. 712–720, 2020.
- Adnan Darwiche and Chunxi Ji. On the Computation of Necessary and Sufficient Explanations. In *Proc. 36th AAAI Conf. on Artificial Intelligence*, pp. 5582–5591, 2022.
- Sanjoy Dasgupta, Nave Frost, and Michal Moshkovitz. Framework for Evaluating Faithfulness of Local Explanations. In *Proc. 39th Int. Conf. on Machine Learning (ICML)*, pp. 4794–4815, 2022.
- Ian Davidson, Nicolás Kennedy, and SS Ravi. CXAD: Contrastive Explanations for Anomaly Detection: Algorithms, Complexity Results and Experiments. *Transactions on Machine Learning Research (TMLR)*, 2025.
- Alessandro De Palma, Greta Dolcetti, and Caterina Urban. Faster Verified Explanations for Neural Networks. 2025. Technical Report. <https://arXiv.org/abs/2512.00164>.
- Amit Dhurandhar, Pin-Yu Chen, Ronny Luss, Chun-Chen Tu, Paishun Ting, Karthikeyan Shanmugam, and Payel Das. Explanations Based on the Missing: Towards Contrastive Explanations with Pertinent Negatives. *Proc. 32nd Conf. on Neural Information Processing Systems (NeurIPS)*, 2018.

-
- 702 Hidde Fokkema, Rianne De Heide, and Tim Van Erven. Attribution-Based Explanations that Provide
703 Recourse Cannot be Robust. *Journal of Machine Learning Research (JMLR)*, 24(360):1–37, 2023.
704
- 705 Hidde Fokkema, Damien Garreau, and Tim van Erven. The Risks of Recourse in Binary Classification.
706 In *Proc. 27th Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*, pp. 550–558, 2024.
- 707 Tobias Geibinger, Reijo Jaakkola, Antti Kuusisto, Xinghan Liu, and Miikka Vilander. Why This and
708 Not That? A Logic-based Framework for Contrastive Explanations. In *Proc. 19th European Conf.
709 on Logics in Artificial Intelligence (JELIA)*, pp. 45–60. Springer, 2025.
- 710 Joshua Größer and Ostap Okhrin. Copulae: An Overview and Recent Developments. *Wiley Interdis-*
711 *ciplinary Reviews: Computational Statistics*, 14(3):e1557, 2022.
712
- 713 Riccardo Guidotti. Counterfactual Explanations and How to Find Them: Literature Review and
714 Benchmarking. *Data Mining and Knowledge Discovery (DMKD)*, 38(5):2770–2824, 2024.
- 715 Hyukjun Gweon, Matthias Schonlau, and Stefan H Steiner. The K Conditional Nearest Neighbor
716 Algorithm for Classification and Class Probability Estimation. *PeerJ Computer Science*, 5:e194,
717 2019.
- 718 Itamar Hadad, Shahaf Bassan, and Guy Katz. Formal Mechanistic Interpretability: Automated Circuit
719 Discovery with Provable Guarantees. In *Proc. 14th Int. Conf. on Learning Representations (ICLR)*,
720 2026.
721
- 722 Ouns El Harzli, Bernardo Cuenca Grau, and Ian Horrocks. Cardinality-Minimal Explanations for
723 Monotonic Neural Networks. In *Proc. 32nd Int. Joint Conf. on Artificial Intelligence (IJCAI)*, pp.
724 3677–3685, 2023.
- 725 W Keith Hastings. Monte Carlo Sampling Methods using Markov Chains and their Applications.
726 *Biometrika*, 57:97–109, 1970.
727
- 728 Rongjie Huang, Max W. Y. Lam, Jun Wang, Dan Su, Dong Yu, Yi Ren, and Zhou Zhao. FastDiff: A
729 Fast Conditional Diffusion Model for High-Quality Speech Synthesis. In *Proc. 31st Int. Joint Conf.
730 on Artificial Intelligence (IJCAI)*, pp. 4157–4163, 2022.
- 731 Xuanxiang Huang, Yacine Izza, Alexey Ignatiev, and Joao Marques-Silva. On Efficiently Explaining
732 Graph-Based Classifiers. In *Proc. 18th Int. Conf. on Principles of Knowledge Representation and
733 Reasoning (KR)*, 2021.
- 734 Alexey Ignatiev, Nina Narodytska, and Joao Marques-Silva. Abduction-Based Explanations for
735 Machine Learning Models. In *Proc. 33rd AAAI Conference on Artificial Intelligence*, pp. 1511–
736 1519, 2019.
- 737 Alexey Ignatiev, Martin Cooper, Mohamed Siala, Emmanuel Hebrard, and Joao Marques-Silva.
738 Towards Formal Fairness in Machine Learning. In *Proc. 26th Int. Conf. on Principles and Practice
739 of Constraint Programming (CP)*, pp. 846–867, 2020a.
- 740 Alexey Ignatiev, Nina Narodytska, Nicholas Asher, and Joao Marques-Silva. From Contrastive to
741 Abductive Explanations and Back Again. In *Proc. Int. Conf. of the Italian Association for Artificial
742 Intelligence (AIxAI)*, pp. 335–355, 2020b.
- 743 Alexey Ignatiev, Yacine Izza, Peter J Stuckey, and Joao Marques-Silva. Using MaxSAT for Efficient
744 Explanations of Tree Ensembles. In *Proc. 36th AAAI Conference on Artificial Intelligence*, pp.
745 3776–3785, 2022.
746
- 747 Rishabh K Iyer and Jeff A Bilmes. Submodular Optimization with Submodular Cover and Submodular
748 Knapsack Constraints. *Proc. 27th Conf. on Neural Information Processing Systems (NeurIPS)*,
749 2013.
750
- 751 Yacine Izza and Joao Marques-Silva. On Explaining Random Forests with SAT. In *Proc. 30th Int.
752 Joint Conf. on Artificial Intelligence (IJCAI)*, 2021.
753
- 754 Yacine Izza, Xuanxiang Huang, Alexey Ignatiev, Nina Narodytska, Martin Cooper, and Joao Marques-
755 Silva. On Computing Probabilistic Abductive Explanations. *Int. Journal of Approximate Reasoning
(IJAR)*, 159:108939, 2023.

-
- Yacine Izza, Xuanxiang Huang, Antonio Morgado, Jordi Planes, Alexey Ignatiev, and Joao Marques-Silva. Distance-Restricted Explanations: Theoretical Underpinnings & Efficient implementation. In *Proc. 21st Int. Conf. on Principles of Knowledge Representation and Reasoning (KR)*, pp. 475–486, 2024a.
- Yacine Izza, Kuldeep S Meel, and Joao Marques-Silva. Locally-Minimal Probabilistic Explanations. In *Proc. 27th European Conf. on Artificial Intelligence (ECAI)*, 2024b.
- Reijo Jaakkola, Tomi Janhunen, Antti Kuusisto, Masood Feyzbakhsh Rankooh, and Miikka Vilander. Explainability via Short Formulas: the Case of Propositional Logic with Implementation. *Journal of Artificial Intelligence Research (JAIR)*, 83, 2025.
- Alon Jacovi, Swabha Swayamdipta, Shauli Ravfogel, Yanai Elazar, Yejin Choi, and Yoav Goldberg. Contrastive Explanations for Model Interpretability. In *Proc. Conf. Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 1597–1611, 2021.
- Helen Jin, Anton Xue, Weiqiu You, Surbhi Goel, and Eric Wong. Probabilistic Stability Guarantees for Feature Attributions. In *Proc. 39th Conf. on Neural Information Processing Systems (NeurIPS)*, 2025.
- Amir-Hossein Karimi, Bernhard Schölkopf, and Isabel Valera. Algorithmic recourse: from counterfactual explanations to interventions. In *Proc. 4th ACM Conf. on Fairness, Accountability, and Transparency (FAccT)*, pp. 353–362, 2021.
- Eoin M Kenny and Mark T Keane. On Generating Plausible Counterfactual and Semi-Factual Explanations for Deep Learning. In *Proc. 35th AAAI Conf. on Artificial Intelligence*, number 13, pp. 11575–11585, 2021.
- Alexander Kozachinskiy. Inapproximability of Sufficient Reasons for Decision Trees. 2023. Technical Report. <https://arXiv.org/abs/2304.02781>.
- I Elizabeth Kumar, Suresh Venkatasubramanian, Carlos Scheidegger, and Sorelle Friedler. Problems with Shapley-Value-Based Explanations as Feature Importance Measures. In *Proc. 37th Int. Conf. on Machine Learning (ICML)*, pp. 5491–5500, 2020.
- Emanuele La Malfa, Agnieszka Zbrzezny, Rhiannon Michelmore, Nicola Paoletti, and Marta Kwiatkowska. On Guaranteed Optimal Robust Explanations for NLP Models. In *Proc. Int. Joint Conf. on Artificial Intelligence (IJCAI)*, pp. 2658–2665, 2021.
- Eduardo Sany Laber. The Computational Complexity of some Explainable Clustering Problems. *Information Processing Letters*, 184:106437, 2024.
- Shuai Li, Yingjie Zhang, Hongtu Zhu, Christina Wang, Hai Shu, Ziqi Chen, Zhuoran Sun, and Yanfeng Yang. K-Nearest-Neighbor Local Sampling Based Conditional Independence Testing. *Proc. 37th Conf. on Neural Information Processing Systems (NeurIPS)*, pp. 23321–23344, 2023.
- Gianluigi Lopardo, Frédéric Precioso, and Damien Garreau. A Sea of Words: An In-Depth Analysis of Anchors for Text Data. In *Proc. 26th Int. Conf. on Artificial Intelligence and Statistics (AISTATS) 2023*, 2023.
- Scott M Lundberg and Su-In Lee. A Unified Approach to Interpreting Model Predictions. *Proc. 31st Conf. on Neural Information Processing Systems (NeurIPS)*, 2017.
- Joao Marques-Silva and Alexey Ignatiev. Delivering Trustworthy AI through formal XAI. In *Proc. 36th AAAI Conf. on Artificial Intelligence*, pp. 12342–12350, 2022.
- Joao Marques-Silva and Alexey Ignatiev. No Silver Bullet: Interpretable ML Models Must be Explained. *Frontiers in Artificial Intelligence*, 6:1128212, 2023.
- Joao Marques-Silva, Thomas Gerspacher, Martin Cooper, Alexey Ignatiev, and Nina Narodytska. Explaining Naive Bayes and Other Linear Classifiers with Polynomial Time and Delay. In *Proc. 33rd Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp. 20590–20600, 2020.

-
- Joao Marques-Silva, Thomas Gerspacher, Martin Cooper, Alexey Ignatiev, and Nina Narodytska. Explanations for Monotonic Classifiers. In *Proc. Int. Conf. on Machine Learning (ICML)*, pp. 7469–7479, 2021.
- Reda Marzouk and Colin De La Higuera. On the Tractability of SHAP Explanations under Markovian Distributions. In *Proc. 41st Int. Conf. on Machine Learning (ICML)*, pp. 34961–34986, 2024.
- Reda Marzouk, Shahaf Bassan, and Guy Katz. SHAP Meets Tensor Networks: Provably Tractable Explanations with Parallelism. In *Proc. 39th Conf. on Neural Information Processing Systems (NeurIPS)*, 2025a.
- Reda Marzouk, Shahaf Bassan, Guy Katz, and De la Higuera. On the Computational Tractability of the (Many) Shapley Values. In *Proc. 28th Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*, 2025b.
- Kuldeep S Meel, Aditya A Shrotri, and Moshe Y Vardi. Not all FPRASs are Equal: Demystifying FPRASs for DNF-Counting. *Constraints*, 24(3):211–233, 2019.
- Ramaravind K Mothilal, Amit Sharma, and Chenhao Tan. Explaining Machine Learning Classifiers Through Diverse Counterfactual Explanations. In *Proc. 3rd Conf. on Fairness, Accountability, and Transparency (FAccT)*, pp. 607–617, 2020.
- Vidya Muthukumar, Tejaswini Pedapati, Nalini Ratha, Prasanna Sattigeri, Chai-Wah Wu, Brian Kingsbury, Abhishek Kumar, Samuel Thomas, Aleksandra Mojsilovic, and Kush R Varshney. Understanding Unequal Gender Classification Accuracy from Face Images. 2018. Technical Report. <https://arXiv.org/abs/1812.00099>.
- Sebastian Ordyniak, Giacomo Paesani, and Stefan Szeider. The Parameterized Complexity of Finding Concise Local Explanations. In *Proc. 32nd Int. Joint Conf. on Artificial Intelligence (IJCAI)*, 2023.
- Sebastian Ordyniak, Giacomo Paesani, Mateusz Rychlicki, and Stefan Szeider. Explaining Decisions in ML Models: A Parameterized Complexity Analysis. In *Proc. 21st Int. Conf. on Principles of Knowledge Representation and Reasoning (KR)*, pp. 563–573, 2024.
- Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. Anchors: High-Precision Model-Agnostic Explanations. In *Proc. 32nd AAAI Conf. on Artificial Intelligence*, 2018.
- Majun Shi, Zishen Yang, and Wei Wang. Minimum Non-Submodular Cover Problem with Applications. *Applied Mathematics and Computation*, 410:126442, 2021.
- Andy Shih, Arthur Choi, and Adnan Darwiche. A Symbolic Approach to Explaining Bayesian Network Classifiers. In *Proc. 27th Int. Joint Conf. on Artificial Intelligence (IJCAI)*, pp. 5103–5111, 2018.
- Bernardo Subercaseaux, Marcelo Arenas, and Kuldeep S Meel. Probabilistic Explanations for Linear Models. In *Proc. 39th AAAI Conf. on Artificial Intelligence*, pp. 20655–20662, 2025.
- Guy Van den Broeck, Anton Lykov, Maximilian Schleich, and Dan Suciu. On the Tractability of SHAP Explanations. *Journal of Artificial Intelligence Research (JAIR)*, 74:851–886, 2022.
- Gal Vardi, Gilad Yehudai, and Ohad Shamir. On the Optimal Memorization Power of ReLU Neural Networks. In *Proc. 10th Int. Conf. on Learning Representations, (ICLR)*, 2022.
- Vijay V Vazirani. *Approximation Algorithms*. Springer, 2001.
- Sahil Verma, Varich Boonsanong, Minh Hoang, Keegan Hines, John Dickerson, and Chirag Shah. Counterfactual Explanations and Algorithmic Recourses for Machine Learning: A Review. *ACM Computing Surveys*, 56(12):1–42, 2024.
- Stephan Wäldchen, Jan Macdonald, Sascha Hauch, and Gitta Kutyniok. The Computational Complexity of Understanding Binary Classifier Decisions. *Journal of Artificial Intelligence Research (JAIR)*, 70:351–387, 2021.
- E. Wang, P. Khosravi, and G. Van den Broeck. Probabilistic Sufficient Explanations. In *Proc. 30th Int. Joint Conf. on Artificial Intelligence (IJCAI)*, 2021a.

-
- 864 S. Wang, H. Zhang, K. Xu, X. Lin, S. Jana, C. Hsieh, and Z. Kolter. Beta-Crown: Efficient Bound
865 Propagation with Per-Neuron Split Constraints for Neural Network Robustness Verification. In
866 *Proc. 35th Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp.
867 29909–29921, 2021b.
- 868 Greta Warren, Eoin Delaney, Christophe Guéret, and Mark T Keane. Explaining Multiple Instances
869 Counterfactually: User Tests of Group-Counterfactuals for XAI. In *Proc. 32nd Int. Conf. on*
870 *Case-Based Reasoning (ICCBR)*, pp. 206–222, 2024.
- 871 Ingo Wegener. The Complexity of Symmetric Boolean Functions. *Computation Theory and Logic*,
872 pp. 433–442, 2005.
- 873 Laurence A Wolsey. An Analysis of the Greedy Algorithm for the Submodular Set Covering Problem.
874 *Combinatorica*, 2(4):385–393, 1982.
- 875 Haoze Wu, Omri Isac, Aleksandar Zeljić, Teruhiro Tagomori, Matthew Daggitt, Wen Kokke, Idan
876 Refaeli, Guy Amir, Kyle Julian, Shahaf Bassan, et al. Marabou 2.0: A Versatile Formal Analyzer
877 of Neural Networks. In *Proc. 36th Int. Conf. on Computer Aided Verification (CAV)*, pp. 249–264,
878 2024a.
- 879 Min Wu, Haoze Wu, and Clark Barrett. Verix: Towards Verified Explainability of Deep Neural
880 Networks. In *Proc. 36th Int. Conf. on Advances in Neural Information Processing Systems*
881 *(NeurIPS)*, 2024b.
- 882 Lei Xu, Maria Skoularidou, Alfredo Cuesta-Infante, and Kalyan Veeramachaneni. Modeling Tabular
883 Data Using Conditional GAN. *Proc. 33rd Conf. on Neural Information Processing Systems*
884 *(NeurIPS)*, 2019.
- 885 Anton Xue, Rajeev Alur, and Eric Wong. Stability Guarantees for Feature Attributions with Multi-
886 plicative Smoothing. *Proc. 37th Conf. on Neural Information Processing Systems (NeurIPS)*, pp.
887 62388–62413, 2023.
- 888 Kayo Yin and Graham Neubig. Interpreting Language Models with Contrastive Explanations. In
889 *Proc. Conf. on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 184–198, 2022.
- 890 Jinqiang Yu, Alexey Ignatiev, Peter Stuckey, Nina Narodytska, and Joao Marques-Silva. Eliminating
891 the Impossible, Whatever Remains Must Be True: On Extracting and Applying Background
892 Knowledge in the Context of Formal Explanations. In *Proc. 37th AAAI Conf. on Artificial*
893 *Intelligence*, pp. 4123–4131, 2023.
- 894
895
896
897
898
899
900
901
902
903
904
905
906
907
908
909
910
911
912
913
914
915
916
917

Appendix

The appendix contains formalizations and proofs that were mentioned throughout the paper:

Appendix A contains extended related work on Formal XAI.

Appendix B contains the formalizations of the models and distributions used in this work.

Appendix C contains the proof of Proposition 1.

Appendix D contains the proof of Proposition 2.

Appendix E contains the proof of Proposition 3.

Appendix F contains the proof of Proposition 4.

Appendix G contains the proof of Proposition 5.

Appendix H contains the proof of Theorem 1.

Appendix I contains the proof of Proposition 6.

Appendix J contains the proof of Theorem 2.

Appendix K contains the proof of Theorem 3.

Appendix L contains the proof of Theorem 4.

Appendix O contains an LLM usage disclosure.

A EXTENDED RELATED WORK ON FORMAL XAI

Our work falls within the line of research on *formal explainable AI* (formal XAI) (Marques-Silva & Ignatiev, 2022), which studies explanations equipped with provable guarantees (Yu et al., 2023; Darwiche & Ji, 2022; Darwiche, 2023; Shih et al., 2018; Azzolin et al., 2025; Audemard et al., 2021; Calautti et al., 2025). A central theme in this area concerns the computational complexity of generating such guaranteed explanations (Barceló et al., 2020; ?; Marzouk et al., 2025a;b; Marzouk & De La Higuera, 2024; Bassan et al., 2025e; Amir et al., 2024; Blanc et al., 2021; 2022; Jaakkola et al., 2025; Amgoud & Cooper, 2026; Geibinger et al., 2025). Since obtaining formal explanations is often computationally intractable for expressive models such as neural networks (Adolfi et al., 2024; Barceló et al., 2020) and tree ensembles (Ordyniak et al., 2024; Bassan et al., 2025a), much of the literature has focused on more tractable model classes (Marques-Silva & Ignatiev, 2023), including decision trees (Bounia, 2025; Arenas et al., 2022; Bounia, 2024; Bounia & Koriche, 2023), monotonic models (Marques-Silva et al., 2021; Harzli et al., 2023), linear models (Marques-Silva et al., 2020; Subercaseaux et al., 2025), and additive models (Bassan et al., 2026).

Beyond complexity-theoretic investigations, substantial effort has also been devoted to computing formal explanations in practice using automated reasoning tools, such as MaxSAT and MILP solvers for tree ensembles (Audemard et al., 2023; 2022a; Ignatiev et al., 2022; Boumazouza et al., 2021), and neural network verification frameworks (Wang et al., 2021b; Wu et al., 2024a) to derive certified explanations for neural networks (Wu et al., 2024b; Bassan & Katz, 2023; Bassan et al., 2023; Izza et al., 2024a; Hadad et al., 2026; Ignatiev et al., 2019). Additional approaches seek to alleviate the inherent computational burden through abstractions (De Palma et al., 2025; Bassan et al., 2025b; Boumazouza et al., 2026), relaxations such as smoothing (Xue et al., 2023; Anani et al., 2025; Jin et al., 2025), or training-time interventions (Alvarez Melis & Jaakkola, 2018; Bassan et al., 2025c;d).

Finally, within formal XAI, a related line of work studies methods that leverage combinatorial and submodular optimization techniques to improve the computation of *local* probabilistic sufficient reasons (Bounia, 2025; Bounia & Koriche, 2023), particularly for decision tree classifiers. As noted in that literature, as well as in this work, the value-function objective underlying these local explanation variants is not submodular in practice. Nevertheless, greedy algorithms inspired by submodular optimization often serve as effective heuristics and can yield high-quality explanations empirically. In contrast, our work establishes a rigorous theoretical connection between combinatorial and submodular optimization objectives and the *global* variants of sufficient and contrastive reasons. This connection provides formal guarantees and opens the door to a broader range of principled algorithmic implementations.

B MODEL AND DISTRIBUTION FORMALIZATIONS

This section formalizes the model and distribution assumptions referenced throughout the paper. Subsection B.1 specifies the model families used in the complexity results, while Subsection B.2 specifies the distribution regimes under which the probabilistic value functions are evaluated.

B.1 MODEL FORMALIZATIONS

This subsection records the base model types used as representative cases in the paper’s analysis: (i) (axis-aligned) decision trees, (ii) linear classifiers, and (iii) neural networks with ReLU activations.

Decision Trees. We define a decision tree (DT) as a directed acyclic graph that represents a function $f : \mathbb{F} \rightarrow [c]$, where $c \in \mathbb{N}$ denotes the number of classes. The graph encodes the function as follows: (i) Each internal node v is assigned a distinct binary input feature from the set $\{1, \dots, n\}$; (ii) Every internal node v has at most k outgoing edges, each corresponding to a value in $[k]$ assigned to v ; (iii) Along any path α in the decision tree, each variable appears at most once; (iv) Each leaf node is labeled with a class from $[c]$. Thus, assigning values to the inputs $\mathbf{x} \in \mathbb{F}$ uniquely determines a path α from the root to a leaf in the DT, where the function output $f(\mathbf{x})$ corresponds to a class label $i \in [c]$. The size of the DT, denoted $|f|$, is defined as the total number of edges in the graph. To allow for flexible modeling, the ordering of input variables $\{1, \dots, n\}$ may differ across distinct paths α and α' , ensuring that no variable is repeated along a single path.

Neural Networks. We present our *hardness* proofs for neural networks with ReLU activations, though our *tractability results* (i.e., polynomial-time algorithms) apply to any architecture that allows for polynomial-time inference — a standard assumption. Thus, all separation results between tractable and intractable cases we prove carry over to *any neural architecture* at least as expressive as a standard feed-forward ReLU network, encompassing many widely used models such as ResNets, CNNs, Transformers, Diffusion models, and more. Moreover, note that any ReLU network can be represented as a fully connected network by assigning zero weights and biases to missing connections. Following standard conventions (Barceló et al., 2020; Bassan et al., 2024; Adolphi et al., 2025), we thus assume the network is fully connected. Specifically, our analysis applies to multi-layer perceptrons (MLPs). Formally, an MLP f consists of $t - 1$ *hidden layers* ($g^{(j)}$ for $j = 1, \dots, t - 1$) and one output layer $g^{(t)}$, where each layer is defined recursively as:

$$g^{(j)} := \sigma^{(j)}(g^{(j-1)}W^{(j)} + b^{(j)}) \quad (9)$$

where $W^{(j)}$ denotes the weight matrix, $b^{(j)}$ the bias vector, and $\sigma^{(j)}$ the activation function of the j -th layer. Accordingly, the model comprises t weight matrices ($W^{(1)}, \dots, W^{(t)}$), t bias vectors ($b^{(1)}, \dots, b^{(t)}$), and t activation functions ($\sigma^{(1)}, \dots, \sigma^{(t)}$).

The input layer is defined as $g^{(0)} = \mathbf{x} \in \{0, 1\}^n$, representing the binary input vector. The dimensions of the network are governed by a sequence of positive integers $d_0, \dots, d_t$, with weight matrices and bias vectors given by $W^{(j)} \in \mathbb{Q}^{d_{j-1} \times d_j}$ and $b^{(j)} \in \mathbb{Q}^{d_j}$, respectively. These parameters are learned during training. Since the model functions as a binary classifier over n features, we set $d_0 = n$ and $d_t = 1$. The hidden layers use the ReLU activation function, defined by $\text{ReLU}(x) = \max(0, x)$. Although a sigmoid activation is typically used during training, for interpretability purposes, we assume the output layer applies a threshold-based step function, defined as $\text{step}(\mathbf{z}) = 1$ if $\mathbf{z} \geq 0$ and $\text{step}(\mathbf{z}) = 0$ otherwise.

Tree Ensembles. Many popular ensemble methods exist, but since our aim is to provide post-hoc explanations, we focus on the *inference* phase rather than the training process. Our analysis centers on ensemble families that rely on either *majority voting* (e.g., Random Forests) or *weighted voting* (e.g., XGBoost) during inference. However, as with our treatment of neural networks, our *tractability results* — namely, polynomial-time algorithms — extend to any possible ensemble configuration with polynomial-time inference, encompassing an even broader range of ensemble configurations.

Majority Voting Inference. In *majority voting* inference, the final prediction $f(\mathbf{x})$ is assigned to the class $j \in [c]$ that receives the majority of votes among the individual predictions $f_i(\mathbf{x})$ from all $i \in [k]$ (i.e., from each tree in the ensemble).

$$f(\mathbf{x}) := \text{majority}(\{f_i(\mathbf{x}) \mid i \in [k]\}) \quad (10)$$

where $\text{majority}(S)$ denotes the most frequent label in the multiset S . If there is a tie, it is resolved by a fixed tie-breaking rule (e.g., lexicographic order or predefined priority).

Weighted Voting Inference. In *weighted voting* inference, each model in the ensemble is assigned a weight $\phi_i \in \mathbb{Q}$ representing its relative importance. The predicted class is the one with the highest total weight across all models. Formally, for any $\mathbf{x} \in \mathbb{F}$, we define f as:

$$f(\mathbf{x}) := \arg \max_{j \in [c]} \sum_{i=1}^k \phi_i \cdot \mathbb{I}[f_i(\mathbf{x}) = j] \quad (11)$$

where $\mathbb{I}$ denotes the identity function.

B.2 DISTRIBUTION FORMALIZATIONS

This subsection makes explicit the distribution definitions discussed in the main paper.

Empirical Distributions. An empirical distribution represents $\mathcal{D}$ through a finite dataset $\mathbf{D}$ containing inputs $\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}$. Under this representation, the probability of an input $\mathbf{x} \in \mathbb{F}$ is its frequency in $\mathbf{D}$, namely:

$$\Pr(\mathbf{x}) := \frac{1}{|\mathbf{D}|} \sum_{i=1}^{|\mathbf{D}|} \mathbb{I}(\mathbf{z}^i = \mathbf{x}) \quad (12)$$

Independent Distributions. Formally, given a probability value $p(\mathbf{x}_i) \in [0, 1]$ defined for each individual input feature, we say that the distribution $\mathcal{D}$ is independent if the joint probability over inputs is given by $\Pr(\mathbf{x}) := \prod_{i \in [n]} p(\mathbf{x}_i)$.

We note that when $p(\mathbf{x}_i) = p(\mathbf{x}_j)$ holds for all $i, j \in [n]$, the distribution reduces to the *uniform distribution*, a special case of independent distributions.

General Distributions. Many proofs in the paper — particularly those establishing structural properties of value functions — are stated for general distribution assumptions rather than a single data representation. Empirical distributions are common in XAI when the training data are used as a proxy, but other approximation frameworks include k-NN resampling from nearby points (Li et al., 2023; Almanjahie et al., 2018; Gweon et al., 2019), copulas for tabular dependencies (Größer & Okhrin, 2022), and conditional generative models such as CTGAN (Xu et al., 2019) or conditional diffusion models (Huang et al., 2022). These choices also appear in counterfactual explanation and algorithmic recourse (Karimi et al., 2021; Fokkema et al., 2024; 2023; Verma et al., 2024), which are closely related to contrastive explanations. When a value function is not computed directly from model structure (e.g., leaf enumeration in decision trees) or from an empirical distribution, it may instead be approximated through sampling methods such as Monte Carlo (Hastings, 1970), following approaches similar to (Subercaseaux et al., 2025; Lopardo et al., 2023).

B.3 SUBSET VS. CARDINAL MINIMALITY

In this subsection, we give a more detailed discussion of the distinction between cardinal and subset minimality. Cardinal minimality offers a substantially stronger guarantee, as it corresponds to a globally minimal explanation size, whereas subset minimality only ensures a local form of minimality.

To clarify why cardinal minimality is stronger, note that if $S \subseteq [n]$ is a cardinally minimal explanation, then no smaller set S' with $|S'| < |S|$ can qualify as an explanation. Hence, no strict subset $S' \subsetneq S$ is an explanation, which means S is also subset minimal.

The converse, however, does not hold. Consider the function:

$$f := \mathbf{x}_1 \vee (\mathbf{x}_2 \wedge \mathbf{x}_3) \quad (13)$$

and the assignment $\mathbf{x} := (1, 1, 1)$, which gives $f(1, 1, 1) = 1$. Fixing feature $\mathbf{x}_1 = 1$ yields a cardinally minimal (and subset minimal) sufficient reason, since the prediction remains 1 regardless of $\mathbf{x}_2, \mathbf{x}_3$. But fixing both $\mathbf{x}_2 = 1$ and $\mathbf{x}_3 = 1$ also gives a subset minimal explanation—yet not a cardinally minimal one. Thus, while every cardinally minimal explanation is subset minimal, not every subset minimal explanation is cardinally minimal.

C PROOF OF PROPOSITION 1

Proposition 1. *While the non-probabilistic case ($\delta = 1$) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary δ), there exist functions f that have $\Theta(\frac{2^n}{\sqrt{n}})$ subset-minimal global sufficient reasons.*

Proof. It is known that certain Boolean functions admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons for some input $\mathbf{x}$ Bassan et al. (2024). In particular, this phenomenon was shown to occur in functions of the following form. We demonstrate that the same function admits an exponential number of *global* and *probabilistic* sufficient reasons. Notably, the mentioned function is a special case of a broader class of *threshold Boolean functions* described in Wegener (2005). Specifically, for $n = 2k + 1$ with $k \in \mathbb{N}$, this function is defined as:

$$f(\mathbf{x}) := \begin{cases} 1 & \text{if } \sum_{i=1}^n \mathbf{x}_i \geq k + 1 \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

where $\mathbf{x}$ is drawn from a uniform distribution $\mathcal{D}$. Notably, the function f is *symmetric*, in the sense of symmetric threshold Boolean functions Wegener (2005): its output depends solely on the number of 1’s (or, equivalently, 0’s) in the input and is invariant under any permutation of input bits. Furthermore, the two extreme inputs — the all-zeros vector $(0, \dots, 0)$ and the all-ones vector $(1, \dots, 1)$ — each admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons.

Since the function f is symmetric, the condition $v_{\text{diff}}^g(S_1) = v_{\text{diff}}^g(S_2)$ holds for any two subsets $S_1, S_2 \subseteq [n]$. Moreover, it can be verified that for any subset $S \subseteq [n]$ of size $k + 1$, we have:

$$(\forall i \in S). \quad v_{\text{diff}}^g(S) > v_{\text{diff}}^g(S \setminus \{i\}) \quad (15)$$

We also deliberately set δ to satisfy:

$$v_{\text{diff}}^g(S \setminus \{i\}) < \delta \leq v_{\text{diff}}^g(S), \quad (16)$$

and also set $k := \lfloor \frac{n}{2} \rfloor$. Each of these subsets is *subset-minimal* because any subset of size at most $\lfloor \frac{n}{2} \rfloor - 1$ fails to be a sufficient reason for $\langle f, \mathbf{x} \rangle$ (with respect to the δ threshold). Therefore, we can directly apply the same analysis as in (Bassan et al., 2024). In particular, there are exactly $\binom{n}{\lfloor \frac{n}{2} \rfloor}$ subset-minimal local sufficient reasons for $\langle f, \mathbf{x} \rangle$. Using Stirling’s approximation, we obtain:

$$\lim_{n \rightarrow \infty} \frac{2\sqrt{2\pi}}{e^2} \cdot \frac{2^n}{\sqrt{n}} \leq \binom{n}{\lfloor \frac{n}{2} \rfloor} \leq \frac{e}{\pi} \cdot \frac{2^n}{\sqrt{n}} \quad (17)$$

This yields the corresponding bound on the number of subset-minimal global δ -sufficient reasons.

□

D PROOF OF PROPOSITION 2

Proposition 2. *While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{diff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{diff}^g are monotonic non-decreasing.*

Proof. This proof establishes monotonicity for both global explanation types — sufficient and contrastive. We start with global sufficiency. To build intuition, we first handle Boolean functions, then extend the argument to discrete multi-valued input domains and multiple output classes. We further extend the result to continuous input domains and show that the monotonicity property holds for any well-defined classification function. Finally, we demonstrate how the same monotonicity guarantees apply to global contrastive explanations. Importantly, none of our proofs rely on the assumption of feature independence — the monotonicity property holds strongly for *any* underlying distribution.

Finally, we present a simple example showing that the monotonicity property does not hold for the local probabilistic sufficient or contrastive value functions.

Lemma 1. *The global sufficient value function v_{suff}^g is monotonic non-decreasing for Boolean functions, under any data distribution.*

Proof. Given an arbitrary set $S \in [n]$, and a feature $i \notin S$, we have that:

$$\Pr(\mathbf{x}) = \Pr(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) = \Pr(\mathbf{x}_{\bar{S}} \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \quad (18)$$

Consequently, this implies that:

$$\begin{aligned} \Pr(\mathbf{x}) &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_{S \cup \{i\}}) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_S, x_i) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(x_i \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \end{aligned} \quad (19)$$

Moreover, for any two points $\mathbf{x}, \mathbf{x}' \sim \mathcal{D}$ such that $\mathbf{x}_S = \mathbf{x}'_S$ and $f(\mathbf{x}) = 1 - f(\mathbf{x}')$, it holds that:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) = 1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}') \mid \mathbf{z}_S = \mathbf{x}'_S) \quad (20)$$

To simplify notation, we occasionally omit the distribution notation $\sim \mathcal{D}$ in the local probability expression $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$, and use f^+ and f^- to denote the events $f(\mathbf{z}) = 1$ and $f(\mathbf{z}) = 0$, respectively.

We begin with a technical simplification of the probability term to facilitate the proof, and then proceed to establish monotonicity.

Simplifying the probability term. Let $\mathcal{D}_S$ denote the distribution restricted to the set S . Given a point $\mathbf{x} \sim \mathcal{D}$ and a set S , consider all points $\mathbf{z} \sim \mathcal{D}$ such that $\mathbf{z}_S = \mathbf{x}_S$ and $f(\mathbf{z}) = b$, where $b \in \{0, 1\}$ — they all share the same local probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b \mid \mathbf{z}_S = \mathbf{x}_S)$. For every $\mathbf{x} \sim \mathcal{D}$ that shares the same $\mathbf{x}_S$ and the same output b , we marginalize over $\mathbf{x}_{\bar{S}}$. This yields $\Pr(f(\mathbf{x}) = b, \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S)$, from which we can infer:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b \mid \mathbf{z}_S = \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b \mid \mathbf{x}_S). \quad (21)$$

Thus, for $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$, we get that:

$$\begin{aligned} &\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^+} \Pr(f^+, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S) + \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^-} \Pr(f^-, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^- \mid \mathbf{z}_S = \mathbf{x}_S) \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2 + (\Pr_{\mathbf{z} \sim \mathcal{D}}(f^- \mid \mathbf{z}_S = \mathbf{x}_S))^2] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2 + (1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2], \end{aligned} \quad (22)$$

From which it follows that:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{\mathbf{x}_{S \cup \{t\}} \sim \mathcal{D}_{S \cup \{t\}}} \Pr(\mathbf{x}_{S \cup \{t\}}) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2] \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(x_t = 1, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2] + \\
&\quad \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(x_t = 0, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2]
\end{aligned} \tag{23}$$

Proving monotonicity. We next proceed to prove the monotonicity claim, building on the previous simplification. We start by introducing a few additional notations. Let $g_1^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1)$ and $g_0^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0)$. Similarly, define $P_1 := \Pr(z_t = 1 \mid \mathbf{x}_S)$ and $P_0 := \Pr(z_t = 0 \mid \mathbf{x}_S)$. Then, the following expectations $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ and $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})]$ can be simplified accordingly¹:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 + (1 - (P_1 \cdot g_1^+ + P_0 \cdot g_0^+))^2] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1].
\end{aligned} \tag{24}$$

As a result, we obtain the following:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot ((g_1^+)^2 + (1 - g_1^+)^2) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot ((g_0^+)^2 + (1 - g_0^+)^2) \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1) \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2P_1 \cdot ((g_1^+)^2 - g_1^+) + 2P_0 \cdot ((g_0^+)^2 - g_0^+) + 1] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1].
\end{aligned} \tag{25}$$

Let $\Delta_t(S, f)$ denote $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$. Without loss of generality, we assume that both P_1 and P_0 are positive. Otherwise, it can be verified that $\Delta_t(S, f) = 0$. Then, we get:

¹Note that the notations P_1 , P_0 , g_1^+ , and g_0^+ —as well as all others used throughout the paper—are not fixed constants, but rather depend on the specific input $\mathbf{x}_S$. Formally, they should be written as $P_1(\mathbf{x}_S)$, $P_0(\mathbf{x}_S)$, and so on. However, for readability, we omit the explicit dependence on $\mathbf{x}_S$.

$$\begin{aligned}
\Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot [(P_1 - P_1^2)(g_1^+)^2 + (P_0 - P_0^2)(g_0^+)^2 - 2 \cdot P_1 \cdot P_0 \cdot g_1^+ \cdot g_0^+] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 \cdot P_0 \cdot (g_1^+ - g_0^+)^2 \\
&\geq 0.
\end{aligned} \tag{26}$$

Note that $P_1 - P_1^2 = P_1(1 - P_1) = P_1 P_0$ and similarly: $P_0 - P_0^2 = P_0(1 - P_0) = P_0 P_1$. Using these identities along with the previous result, we conclude that $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ is monotone, thereby concluding our proof. $\square$

Lemma 2. *The global sufficient value function v_{uff}^g is monotonic non-decreasing for functions with discrete multi-valued input domains and multiple output classes, under any data distribution.*

Proof. Extending the proof from the simplified Boolean case, we now generalize the result to functions whose input domains and output ranges each consist of a finite set of values. Let $\mathcal{K}$ represent the set of output classes.

We can generalize equation 22 as follows:

$$\begin{aligned}
\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] &= \\
\sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right]
\end{aligned} \tag{27}$$

Given the condition $\mathbf{z}_S = \mathbf{x}_S$, suppose feature t can take r possible values $\{v_1, \dots, v_r\}$. Then, equation 23 can be generalized as follows:

$$\begin{aligned}
&\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{k=1}^r \left[\sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = v_k}} \Pr(\mathbf{x}_S) \cdot \Pr(z_t = v_k \mid \mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k))^2 \right] \right].
\end{aligned} \tag{28}$$

Let P_k denote $\Pr(z_t = v_k \mid \mathbf{x}_S)$, Let g_k^j denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k)$, that is, z_t takes the value v_k and the output class is j . Then, we have

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \tag{29}$$

and

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k=1}^r P_k \cdot \left(\sum_{j \in \mathcal{K}} (g_k^j)^2 \right) \right]$$

Combining these two implications, we can conclude that:

$$\begin{aligned}
\Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k=1}^r P_k \cdot \left(\sum_{j \in \mathcal{K}} (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r (P_k \cdot g_k^j)^2 \right) + \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \sum_{k \neq l} P_k \cdot P_l \cdot (g_k^j)^2 - \sum_{j \in \mathcal{K}} \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k \cdot P_l \cdot (g_k^j - g_l^j)^2 \right] \\
&\geq 0.
\end{aligned} \tag{30}$$

Hence, $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ is monotone, which completes the proof for this setting. $\square$

Lemma 3. *The global sufficient value function v_{diff}^g is monotonic non-decreasing for functions with continuous input domains and multiple output classes, under any data distribution.*

Proof. We extend the monotonicity results to functions where features are defined over continuous domains. One approach is to partition the domain of the continuous features into two disjoint subdomains and then adapt the proofs in lemma 2 by revising the notation accordingly. The core reasoning remains unchanged when transitioning from the discrete to the continuous case—only the notation differs.

Alternatively, we give a direct proof below. Assume all features lie in continuous domains, and let $p(\mathbf{x}_S, \mathbf{x}_{\bar{S}})$ denote the corresponding joint probability density function. Moreover, let $p(x_i)$ denote the marginal probability density function of x_i , and $\mathcal{X}_i$ denote the domain of feature i .

We define marginal density as

$$p(\mathbf{x}_S) := \int_{\mathbf{x}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} p(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) d\mathbf{x}_{\bar{S}}, \tag{31}$$

where $\mathcal{X}_{\bar{S}}$ represents the domain of features in $\bar{S}$.

And the conditional density is defined as:

$$p(x_t \mid \mathbf{x}_S) := \frac{p(\mathbf{x}_S, x_t)}{p(\mathbf{x}_S)}. \tag{32}$$

The local probability of S given $\mathbf{x}$ is defined as

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \frac{\int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot p(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) d\mathbf{z}_{\bar{S}}}{p(\mathbf{x}_S)}, \tag{33}$$

This term can also be decomposed into:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \int_{z_t \in \mathcal{X}_t} p(z_t \mid \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \tag{34}$$

via an additional feature $t \notin S$.

We then modify equation 27 to

$$\begin{aligned}
&\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right] d\mathbf{x}_S,
\end{aligned} \tag{35}$$

while equation 28 is changed to

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\int_{z_t \in \mathcal{X}_t} p(z_t \mid \mathbf{x}_S) \cdot \left(\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S. \end{aligned} \quad (36)$$

For simplicity, let $Y_j(t)$ denote $\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_t)$, and $p_{t|S}$ denote $p(z_t \mid \mathbf{x}_S)$. Note that both $Y_j(t)$ and $p_{t|S}$ are associated with a fixed $\mathbf{x}_S$ ². We prove monotonicity as follows:

$$\begin{aligned} & \Delta_t(S, f) \\ &= \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \left(\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S - \\ & \quad \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \right)^2 \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t)^2 dz_t - \left(\int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t) dz_t \right)^2 \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{2}{2} \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t)^2 - Y_j(t)Y_j(t')) dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &\geq 0. \end{aligned} \quad (37)$$

□

Proposition 2. While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.

Proof. As a consequence of lemmas 1 to 3, we conclude that the global sufficient value function v_{suff}^g is monotonic non-decreasing for any well-defined classification function — whether its features are Boolean, discrete, continuous, or a combination — under any data distribution.

$$v_{\text{con}}^g(S) = 1 - v_{\text{suff}}^g(\bar{S}) \quad (38)$$

This implies that v_{con}^g is monotonic non-increasing with respect to adding features to $\bar{S}$. Equivalently, v_{con}^g is monotonic non-decreasing as features are removed from $\bar{S}$ and added to S , thus completing the proof.

□

E PROOF OF PROPOSITION 3

Proposition 3. While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{suff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{suff}^g is supermodular.

Proof. This section establishes the supermodularity property for both global sufficient and contrastive explanations. As with monotonicity, we begin with Boolean functions and progressively generalize

²Changing $\mathbf{x}_S$ alters their values, we omit $\mathbf{x}_S$ in the notation for brevity.

to multi-valued discrete and continuous input domains, and finally consider the case of well-defined classification functions. Unlike the monotonicity property, however, these proofs require the assumption of feature independence. We also provide a counterexample to show that supermodularity may fail when this assumption is relaxed.

Lemma 4. *The global sufficient value function v_{uff}^g is supermodular for Boolean functions, under the assumption of feature independence.*

Proof. Particularly, we will prove that:

$$\Delta_t(S, f) \leq \Delta_t(S', f), \quad (39)$$

where $S \subseteq S'$ and $t \notin S'$. That is, $\Delta_t(S, f)$ is monotone.

Clearly, if $S' = S$, then $\Delta_t(S, f) \leq \Delta_t(S', f)$ holds. Let $S' = S \cup \{i\}$, where $i \notin S$ and $i \neq t$. For the local probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$, under the assumption of feature independence, we have that the following holds:

$$\begin{aligned} \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) \\ = \Pr(z_i = 1) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1) + \\ \Pr(z_i = 1) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0) + \\ \Pr(z_i = 0) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1) + \\ \Pr(z_i = 0) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0). \end{aligned} \quad (40)$$

Let P_1, P_0, Q_1 , and Q_0 denote $\Pr(z_t = 1), \Pr(z_t = 0), \Pr(z_i = 1)$, and $\Pr(z_i = 0)$, respectively. Moreover, let $g_{11}^+, g_{10}^+, g_{01}^+$, and g_{00}^+ denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1)$, and $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0)$, respectively.

We further decompose equation 26 into:

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [(Q_1 \cdot g_{11}^+ + Q_0 \cdot g_{01}^+) - (Q_1 \cdot g_{10}^+ + Q_0 \cdot g_{00}^+)]^2 \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+) + Q_0 \cdot (g_{01}^+ - g_{00}^+)]^2. \end{aligned} \quad (41)$$

Similarly, the following also holds:

$$\Delta_t(S \cup \{i\}, f) = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+)^2 + Q_0 \cdot (g_{01}^+ - g_{00}^+)^2]. \quad (42)$$

Thus, we get

$$\begin{aligned} \Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot Q_1 Q_0 \cdot (g_{11}^+ - g_{10}^+ - g_{01}^+ + g_{00}^+)^2 \\ &\geq 0. \end{aligned} \quad (43)$$

This implies that $\Delta_t(S, f)$ is monotone, that is, $c(S \cup \{t\}, f) - c(S, f) \leq c(S' \cup \{t\}, f) - c(S', f)$. $\square$

Lemma 5. *The global sufficient value function v_{uff}^g is supermodular for functions with discrete multi-valued input domains and multiple output classes, under the assumption of feature independence.*

Proof. We extend the results of the previous section to multi-valued discrete functions, again with the assumption of feature independence. We use P_k to denote $\Pr(z_t = v_k)$, and $Q_{k'}$ to denote $\Pr(z_i = v_{k'})$. Moreover, we use $g_{k'k}^j$ to denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_i = v_{k'}, z_t = v_k)$. We

decompose equation 30 to

$$\begin{aligned}\Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \left(\sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'k}^j - \sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'l}^j \right)^2 \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \left(\sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j) \right)^2 \right].\end{aligned}\quad (44)$$

Likewise, we have

$$\begin{aligned}\Delta_t(S \cup \{i\}, f) &= \sum_{k'=1}^{r'} \left[\sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_i = v_k}} \Pr(\mathbf{x}_S) \cdot \Pr(z_i = v_k) \cdot \sum_{j \in \mathcal{K}} \left(\sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k'=1}^{r'} Q_{k'} \cdot \sum_{j \in \mathcal{K}} \left(\sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right].\end{aligned}\quad (45)$$

Therefore, we get

$$\begin{aligned}\Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} \sum_{k' < l'} P_k P_l Q_{k'} Q_{l'} \cdot (g_{k'k}^j - g_{k'l}^j - g_{l'k}^j + g_{l'l}^j)^2 \right] \\ &\geq 0.\end{aligned}\quad (46)$$

□

Proposition 3. *While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{diff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{diff}^g is supermodular.*

Proof. We can generalize the supermodularity results established in Lemma 5 to the continuous case, assuming that $\mathcal{D}_p$ satisfies feature independence. One approach is to partition the domain of the continuous features into two disjoint subdomains and then adapt the proofs in Lemma 5 by revising the notation accordingly.

Alternatively, we give a direct proof below. Under this assumption, the marginal and conditional densities simplify as follows:

$$p(\mathbf{x}_S) = \prod_{i \in S} p(x_i), \text{ and } p(x_t | \mathbf{x}_S) = p(x_t). \quad (47)$$

Consequently, the local probability of S conditioned on $\mathbf{x}$ becomes:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S) = \int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot \prod_{i \in \bar{S}} p(z_i) d\mathbf{z}_{\bar{S}}. \quad (48)$$

We let $Y_j(i, t)$ denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S, z_i = x_i, z_t = x_t)$, where $i \notin S, t \notin S$ and $i \neq t$. Evidently,

$$Y_j(i, t) = Y_j(t, i), \text{ and } Y_j(t) = \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i.$$

We have:

$$\begin{aligned}
\Delta_t(S, f) &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i - \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t') dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S,
\end{aligned} \tag{49}$$

where

$$\begin{aligned}
&\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \\
&= \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} \cdot (Y_j(i, t) - Y_j(i, t')) \cdot (Y_j(i', t) - Y_j(i', t')) dz_i dz_{i'} dz_t dz_{t'}.
\end{aligned}$$

Let $S' = S \cup \{i\}$, then we have:

$$\begin{aligned}
\Delta_t(S', f) &= \int_{\mathbf{x}_{S'} \in \mathcal{X}_{S'}} p(\mathbf{x}_{S'}) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_{S'} \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \int_{\mathcal{X}_i} p_i \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) dz_i \right] d\mathbf{x}_S \tag{50} \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \right) \right] d\mathbf{x}_S,
\end{aligned}$$

where

$$\begin{aligned}
&\int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \\
&= \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_{i'} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i dz_{i'}.
\end{aligned}$$

Let $\alpha(i, t, t')$ denote $Y_j(i, t) - Y_j(i, t')$ and $\beta(i', t, t')$ denote $Y_j(i', t) - Y_j(i', t')$, we can infer that

$$\begin{aligned}
\Delta_t(S', f) - \Delta_t(S, f) &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \left[\sum_{j \in \mathcal{K}} \frac{1}{4} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} (\alpha(i, t, t') - \beta(i', t, t'))^2 dz_i dz_{i'} dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&\geq 0.
\end{aligned} \tag{51}$$

Consequently, the global sufficient value function v_{suff}^g retains the supermodularity property for any valid classification function, regardless of whether its features are Boolean, discrete, continuous, or a mix thereof.

We next demonstrate that the local probabilistic sufficient value function is neither submodular nor supermodular. As a simple illustration, fix the uniform distribution on the three variables $\{x_1, x_2, x_3\}$ and define a Boolean function as follows:

$$f(x_1, x_2) = (x_1 \vee x_2) \wedge x_3. \tag{52}$$

Consider the input $\mathbf{x} = (1, 1, 1)$ which is classified as 1. For the local probabilistic sufficient value function v_{suff}^l , we have that the following holds:

$$v_{\text{suff}}^l(\emptyset) = \frac{3}{8}, \quad v_{\text{suff}}^l(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{2\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{3\}) = \frac{3}{4},$$

$$v_{\text{suff}}^l(\{1, 2\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{1, 3\}) = 1, \quad v_{\text{suff}}^l(\{2, 3\}) = 1, \quad v_{\text{suff}}^l(\{1, 2, 3\}) = 1.$$

Since $(v_{\text{suff}}^l(\{1, 2\}) - v_{\text{suff}}^l(\{2\})) < (v_{\text{suff}}^l(\{1\}) - v_{\text{suff}}^l(\emptyset)) < (v_{\text{suff}}^l(\{1, 3\}) - v_{\text{suff}}^l(\{3\}))$, the function v_{suff}^l is neither supermodular nor submodular. $\square$

F PROOF OF PROPOSITION 4

Proposition 4. *While the local probabilistic contrastive setting (for any δ) lacks submodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{con}^l is not submodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{con}^g is submodular.*

Proof. Once we establish that the global sufficient setting is supermodular, it directly follows that the global contrastive setting is submodular. Let $S \subseteq [n]$ be the set of features allowed to change, and suppose $S \subseteq S'$ with $i \notin S'$. To compare the marginal contributions $v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)$ and $v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')$, we equivalently compare $(1 - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}))$ and $(1 - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}'))$. Since $S \subseteq S'$, it follows that $\bar{S}' \subseteq \bar{S}$, and therefore we obtain:

$$(v_{\text{suff}}^g(\bar{S}) - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) \geq (v_{\text{suff}}^g(\bar{S}') - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) \quad (53)$$

Which is equivalent to:

$$(v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)) \geq (v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')). \quad (54)$$

Thus, the global contrastive value function v_{con}^g is submodular, hence concluding this proof.

We will now demonstrate that the local probabilistic contrastive value function is neither submodular nor supermodular. As a simple illustration, fix the uniform distribution on the three variables $\{x_1, x_2, x_3\}$ and define a Boolean function as follows:

$$f(x_1, x_2) := (x_1 \vee x_2) \wedge x_3. \quad (55)$$

Consider the input $\mathbf{x} = (1, 1, 1)$ which is classified as 1. For the local probabilistic contrastive value function v_{con}^l , we have:

$$v_{\text{con}}^l(\emptyset) = 1, \quad v_{\text{con}}^l(\{1\}) = 1, \quad v_{\text{con}}^l(\{2\}) = 1, \quad v_{\text{con}}^l(\{3\}) = \frac{1}{2},$$

$$v_{\text{con}}^l(\{1, 2\}) = \frac{3}{4}, \quad v_{\text{con}}^l(\{1, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{2, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{1, 2, 3\}) = \frac{3}{8}.$$

Since $(v_{\text{con}}^l(\{1, 2\}) - v_{\text{con}}^l(\{2\})) < (v_{\text{con}}^l(\{1, 2, 3\}) - v_{\text{con}}^l(\{2, 3\})) < (v_{\text{con}}^l(\{1\}) - v_{\text{con}}^l(\emptyset))$, the function v_{con}^l is neither submodular nor supermodular. $\square$

In the rest of this section, we establish that the feature independence assumption is not just sufficient but also necessary for supermodularity. We illustrate this with the following counterexample.

Define a Boolean function:

$$f(x_1, x_2) := x_1 \quad (56)$$

For this boolean function, it holds that $f(00) = 0$, $f(01) = 0$, $f(10) = 1$, and $f(11) = 1$. Moreover, define a distribution $\mathcal{D}$ on x_1 and x_2 . Recall that the global sufficient value function is:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x} \sim \mathcal{D}} \mathbf{Pr}(\mathbf{x}) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$$

In addition, if we can show that

$$\Delta_t(S, f) \leq \Delta_t(S', f),$$

where $S \subseteq S'$ and $t \notin S'$, then the global sufficient value function is supermodular. When $S = \emptyset$, we have that the following holds:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) \end{aligned}$$

Which is also equivalent to:

$$\begin{aligned} & \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1)) + \\ & \quad \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1)) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1)) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1)) \\ &= (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1))^2 + (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1))^2 \end{aligned}$$

When $S = \{2\}$, we have that the following condition holds:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 0) + \\ & \quad \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 1) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 0) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 1) \end{aligned}$$

This can also be rearranged as follows:

$$\begin{aligned} & \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \frac{\mathbf{Pr}(x_1 = 0, x_2 = 0)}{\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 0)} + \\ & \quad \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \frac{\mathbf{Pr}(x_1 = 0, x_2 = 1)}{\mathbf{Pr}(x_1 = 0, x_2 = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1)} + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \frac{\mathbf{Pr}(x_1 = 1, x_2 = 0)}{\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 0)} + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \frac{\mathbf{Pr}(x_1 = 1, x_2 = 1)}{\mathbf{Pr}(x_1 = 0, x_2 = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1)} \\ &= \frac{\mathbf{Pr}(x_1 = 0, x_2 = 0)^2 + \mathbf{Pr}(x_1 = 1, x_2 = 0)^2}{\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 0)} + \frac{\mathbf{Pr}(x_1 = 0, x_2 = 1)^2 + \mathbf{Pr}(x_1 = 1, x_2 = 1)^2}{\mathbf{Pr}(x_1 = 0, x_2 = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1)} \end{aligned}$$

In contrast, when $S = \{1\}$, we have the the following holds:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\ & \quad \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1) \end{aligned}$$

This can similarly be rearranged as follows:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} = 1
\end{aligned}$$

Finally, when $S = \{1, 2\}$, we have the following condition:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
& = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 0) + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 1) + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 0) + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 1)
\end{aligned}$$

This can likewise be reordered as follows:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0)} + \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0)}{\Pr(x_1 = 1, x_2 = 0)} + \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 1)} \\
& = 1
\end{aligned}$$

Let $t = 2$. Evidently, we have $\Delta_t(\{1\}, f) = 0$. Next, let us analyse $\Delta_t(\emptyset, f)$. We note that if we assume feature independence, we observe that:

$$\begin{aligned}
& (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1))^2 + (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1))^2 \\
& = \Pr(x_1 = 0)^2 + \Pr(x_1 = 1)^2,
\end{aligned}$$

Additionally, another condition that holds is:

$$\begin{aligned}
& \frac{\Pr(x_1 = 0, x_2 = 0)^2 + \Pr(x_1 = 1, x_2 = 0)^2}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \frac{\Pr(x_1 = 0, x_2 = 1)^2 + \Pr(x_1 = 1, x_2 = 1)^2}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} \\
& = \Pr(x_1 = 0)^2 + \Pr(x_1 = 1)^2.
\end{aligned}$$

This implies $\Delta_t(\emptyset, f) = 0$, that is, $\Delta_t(\emptyset, f) \leq \Delta_t(\{1\}, f)$. However, without the assumption of feature independence, we cannot guarantee $\Delta_t(\emptyset, f) = 0$. For example, let:

$$\Pr(x_1 = 0, x_2 = 0) = P_{00}, \Pr(x_1 = 0, x_2 = 1) = P_{01}$$

and

$$\Pr(x_1 = 1, x_2 = 0) = P_{10}, \Pr(x_1 = 1, x_2 = 1) = P_{11}$$

Then, by simplifying $\Delta_t(\emptyset, f)$ we get that the following holds:

$$\Delta_t(\emptyset, f) = \left[\frac{P_{00}^2 + P_{10}^2}{P_{00} + P_{10}} + \frac{P_{01}^2 + P_{11}^2}{P_{01} + P_{11}} \right] - \left[(P_{00} + P_{01})^2 + (P_{10} + P_{11})^2 \right] \quad (57)$$

We then let $P_{00} = 0.4$, $P_{01} = 0.2$, $P_{10} = 0.1$, and $P_{11} = 0.3$, and get that the following holds:

$$\Delta_t(\emptyset, f) = \left[\frac{(0.4)^2 + (0.1)^2}{0.5} + \frac{(0.2)^2 + (0.3)^2}{0.5} \right] - \left[(0.4 + 0.2)^2 + (0.1 + 0.3)^2 \right] = 0.08. \quad (58)$$

This implies that $\Delta_t(\emptyset, f) > \Delta_t(1, f)$, indicating that — unlike in the monotonicity setting where it was not required — feature independence is a necessary condition for supermodularity.

G PROOF OF PROPOSITION 5

Proposition 5. *Computing Algorithm 1 with the local value functions v_{con}^ℓ and v_{suff}^ℓ does not always converge to a subset minimal δ -sufficient/contrastive reason. However, computing it with the global value functions v_{con}^g or v_{suff}^g necessarily produces subset minimal δ -sufficient/contrastive reasons.*

Proof. We note that this proposition follows directly from our proof that both v_{suff}^g and v_{con}^g are monotone non-decreasing (as established in Proposition 2), along with other relevant conclusions drawn in prior work. In the remainder of this section, we elaborate on how this monotonicity property leads to the stated corollary.

The convergence of Algorithm 1 to a subset-minimal δ global sufficient or contrastive reason with respect to the value functions v_{con}^g and v_{suff}^g follows directly from the monotonicity property, for which prior work (Ignatiev et al., 2019; Wu et al., 2024b; Arenas et al., 2022) showed that this type of greedy algorithm yields a subset-minimal explanation when the underlying value function is monotone non-decreasing. The algorithm halts when, for all $i \in S$, the value $v(S \setminus \{i\})$ drops below δ , ensuring that although $v(S) \geq \delta$ (a maintained invariant), removing any element causes the condition to fail. Due to monotonicity, this implies that no proper subset of S satisfies the condition, guaranteeing that S is subset-minimal.

We next turn to explain why Algorithm 1 does not converge to a subset-minimal solution for the local formulation. We note that it is well known that the local probabilistic sufficient value function (and likewise the local probabilistic contrastive value function) is not monotone Arenas et al. (2022); Izza et al. (2023); Subercaseaux et al. (2025); Izza et al. (2024b). As a simple illustration, fix the uniform distribution on the two binary variables $\{x_1, x_2\}$ and define a Boolean function as follows:

$$f(x_1, x_2) := x_1 \vee x_2. \quad (59)$$

Consider the input $\mathbf{x} = (0, 1)$ which is classified as 1, then:

$$v_{\text{suff}}^\ell(\emptyset) = \frac{3}{4}, \quad v_{\text{suff}}^\ell(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^\ell(\{2\}) = 1, \quad v_{\text{suff}}^\ell(\{1, 2\}) = 1. \quad (60)$$

Because $v_{\text{suff}}^\ell(\emptyset) > v_{\text{suff}}^\ell(\{1\}) < v_{\text{suff}}^\ell(\{1, 2\})$, the function v_{suff}^ℓ is not monotone. Likewise, it can be computed that:

$$v_{\text{con}}^\ell(\emptyset) = 1, \quad v_{\text{con}}^\ell(\{1\}) = 1, \quad v_{\text{con}}^\ell(\{2\}) = \frac{1}{2}, \quad v_{\text{con}}^\ell(\{1, 2\}) = \frac{3}{4}. \quad (61)$$

(For v_{con}^ℓ , the S denotes the set of features that are allowed to vary.) Because $v_{\text{con}}^\ell(\emptyset) > v_{\text{con}}^\ell(\{2\}) < v_{\text{con}}^\ell(\{1, 2\})$, the function v_{con}^ℓ is also non-monotone.

□

H PROOF OF THEOREM 1

Theorem 1. *If f is a decision tree and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time. However, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm exists for computing a local δ -sufficient reason for decision trees even under independent distributions.*

Proof. We start by noting that for decision trees, assuming that $\mathcal{D}$ represents independent distributions, the computation of the value function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (62)$$

Can be carried out in polynomial time using the following procedure. Thanks to the tractability of decision trees, we can iterate over all pairs of leaf nodes, each corresponding to partial assignments

$\mathbf{x}_S$ and $\mathbf{z}_{S'}$. For each such pair, if both leaf nodes yield the same prediction under f , we compute the corresponding term in the expected value $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$ by multiplying the respective feature-wise probabilities over the shared features of the two vectors. Under the feature independence assumption, these probabilities decompose, allowing the full expectation to be computed by summing the contributions of all such matching leaf pairs.

Since all probabilities involved are provided as part of the input and each step involves only polynomial-time operations, the entire procedure runs in polynomial time. This establishes that every step in Algorithm 1 is efficient, and thus the algorithm as a whole runs in polynomial time. Finally, combining this with Lemma 2, which proves that v_{suff}^g is monotone non-decreasing, we conclude that Algorithm 1 converges to a subset-minimal explanation. This is because the algorithm halts when, for all $i \in S$, the value $v(S \setminus \{i\})$ falls below δ , while $v(S) \geq \delta$ is preserved throughout. By monotonicity, this guarantees that no strict subset of S satisfies the condition, ensuring S is indeed subset-minimal.

For the remaining part of the claim, the result follows from (Arenas et al., 2022), which showed that, assuming $P \neq NP$, there is no polynomial-time algorithm for computing a subset-minimal δ -sufficient reason for decision trees. While this was proven specifically under the uniform distribution, the hardness clearly extends to independent distributions, which include this case. Combined, these results establish the complexity separation between the local and global variants, thus concluding our proof.

□

I PROOF OF PROPOSITION 6

Proposition 6. *For any model f , and empirical distribution $\mathcal{D}$ — computing a subset-minimal global δ -sufficient or δ -contrastive reason for f can be done in polynomial time.*

Proof. The computation of the global sufficient probability function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (63)$$

or the computation of the global contrastive probability function

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (64)$$

Can be performed in polynomial time when $\mathcal{D}$ is selected from the class of empirical distributions. This is achieved by iterating over pairs of instances $\mathbf{x}, \mathbf{z}$ within the dataset and running an inference through $f(\mathbf{x})$ and $f(\mathbf{z})$ to compute the expected values at each step. Consequently, determining whether the probability function exceeds or falls below a given threshold δ can also be accomplished in polynomial time. Furthermore, leveraging the proof of non-decreasing monotonicity (Proposition D) of the global probability function, we can utilize algorithm 1 for contrastive reason search or for sufficient reason search. By executing a linear number of polynomial queries, a subset-minimal explanation (whether a subset minimal contrastive reason or a subset minimal sufficient reason) can thus be obtained in polynomial time.

□

J PROOF OF THEOREM 2

Theorem 2. *Assuming f is a neural network or a tree ensemble, and $\mathcal{D}$ is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global δ -sufficient and contrastive reasons. However, unless $P = NP$, there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason or a subset minimal local δ -contrastive reason.*

Proof. The first part of the proof follows directly from Proposition 6, which established that this holds for any model, since both $v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$ and $v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$ can be computed in polynomial time from

the empirical distribution. For the second part, we proceed by proving two lemmas: one establishing the intractability of computing the sufficient case, and the other addressing the contrastive case.

Lemma 6. *Unless $P_{TIME} = NP$, then there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason for either a tree ensemble or a neural network, under empirical distributions.*

Proof. We will first prove this claim for neural networks and then extend the result to tree ensembles. We will establish this claim by demonstrating that if a polynomial-time algorithm exists for computing a subset-minimal local δ sufficient reason for neural networks, then it would enable us to solve the classic NP-complete *CNF-SAT* problem in polynomial time. The *CNF-SAT* problem is defined as follows:

CNF-SAT:

Input: A formula in conjunctive normal form (CNF): ϕ .

Output: Yes, if there exists an assignment to the n literals of ϕ such that ϕ is evaluated to True, and No otherwise

Our proof will also utilize the following lemma (with its proof provided in Barceló et al. (2020)):

Lemma 7. *Any boolean circuit ϕ can be encoded into an equivalent MLP over the binary domain $\{0, 1\}^n \rightarrow \{0, 1\}$ in polynomial time.*

Proof. We will actually establish hardness for a simpler, more specific case, which will consequently imply hardness for the more general setting. In this case, we assume that the empirical distribution $\mathcal{D}$ consists of a single element, which, for simplicity, we take to be the zero vector $\mathbf{0}_n$. Since the sufficiency of S in this scenario depends only on the local instance $\mathbf{x}$ and the single instance $\mathbf{0}_n$ in our empirical distribution, the nature of the input format (whether discrete, continuous, etc.) does not affect the result. Therefore, the hardness results hold universally across all these settings.

Similarly to $\mathbf{0}_n$, we define $\mathbf{1}_n$ as an n -dimensional vector consisting entirely of ones. Given an input ϕ , we initially assign $\mathbf{1}_n$ to ϕ , effectively setting all variables to True. If ϕ evaluates to True under this assignment, then a satisfying truth assignment exists. Therefore, we assume that $\phi(\mathbf{1}_n) = 0$. We next introduce the following definitions:

$$\begin{aligned}\phi_2 &:= (x_1 \wedge x_2 \wedge \dots \wedge x_n), \\ \phi' &:= \phi \vee \phi_2\end{aligned}\tag{65}$$

ϕ' , while no longer a CNF, can still be transformed into an MLP f using Lemma 8, ensuring that f behaves equivalently to ϕ' over the domain $\{0, 1\}^n$. Given our assumption that computing a subset-minimal local δ -sufficient reason is feasible in polynomial time, we can determine one for the instance $\langle f, \mathbf{x} := \mathbf{1}_n \rangle$, $\delta := 1$, noting that the empirical distribution $\mathcal{D}$ is simply defined over the single data point $\mathbf{0}_n$.

We next assert that the subset-minimal δ -sufficient reason generated for $\langle f, \mathbf{x} \rangle$ encompasses the entire input space, i.e., $S = \{1, \dots, n\}$, if and only if $\langle \phi \rangle \notin \text{CNF-SAT}$.

We assume that $S = \{1, \dots, n\}$. Since S is a δ -sufficient reason for $\langle f, \mathbf{x} \rangle$, this simply means that setting the complementary set to any value maintains the prediction. Since the complementary set is $\emptyset$ in this case, this trivially holds. The fact that S is subset-minimal means that any other subset $S' \subseteq S$ satisfies $v(S') < \delta = 1$. Since the probability function $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{x}_S = \mathbf{z}_S)$ is determined by a single point (the distribution contains only the point $\mathbf{0}_n$), the probability function can only take values of 1 or 0. Hence, we also know that $v(S') = 0$. This tells us that, aside from the subset $S = \{1, \dots, n\}$, for any subset $S' \subseteq S$, fixing the features in S' to 1 and the rest to 0 does not result in a classification outcome of 1. Since the ϕ_2 component within ϕ' is True only if all features are assigned 1, this directly implies that ϕ is assigned False for any of these inputs. Since we already know that ϕ does not return a True answer for the vector assignment $\mathbf{1}_n$ (as verified at the beginning), and now we have established that the same holds for all other input vectors, we conclude that $\phi \notin \text{CNF-SAT}$.

Now, suppose that S is a subset that is strictly contained within $\{1, \dots, n\}$. Given that S is sufficient under our definitions of the distribution $\mathcal{D}$ with $\delta = 1$, we can apply the same reasoning as before to

conclude that $v(S) = 1$. This implies that setting the features in S to 1 while setting the remaining features to 0 ensures that the function f evaluates to 1. Since this assignment is necessarily not a vector consisting entirely of ones, it follows that the ϕ_2 component within ϕ' must be False. Consequently, the ϕ component must be True, which implies that $\langle \phi \rangle \in \text{CNF-SAT}$. This completes the proof.

We have established that the following claim holds for neural networks. However, extending the proof to tree ensembles requires a minor and straightforward adaptation. To do so, we will utilize the following lemma, which has been noted in several previous works (Ordyniak et al., 2024; Audemard et al., 2022b):

Lemma 8. *Any CNF or DNF ϕ can be encoded into an equivalent random forest classifier over the binary domain $\{0, 1\}^n \rightarrow \{0, 1\}$ in polynomial time.*

Proof. We note that we can apply the same process used in our proof for neural networks, where we encoded ϕ' into an equivalent neural network. However, ϕ' is no longer a valid CNF due to our construction (though encoding it into an MLP was not an issue, as any Boolean circuit can be transformed into an MLP). Nevertheless, since ϕ' consists of a conjunction of only two terms, we can easily represent it as an equivalent CNF:

$$\phi' := (c_1 \vee \phi_2) \wedge (c_2 \vee \phi_2) \wedge \dots (c_m \vee \phi_2) \quad (66)$$

Where each c_i is a disjunction of a few terms. Consequently, ϕ' is a valid CNF, allowing us to transform it into an equivalent random forest classifier. The reduction we outlined for MLPs applies directly to these models as well, thereby completing the proof for both model families.

□

Lemma 9. *Unless $\text{PTIME} = \text{NP}$, then there is no polynomial time algorithm for computing a subset minimal local δ -contrastive reason for either a tree ensemble or a neural network, under empirical distributions.*

Proof. We will present a proof analogous to the one in Lemma 6. Specifically, we will once again utilize the classical NP-hard CNF-SAT problem defined in Lemma 6. In particular, given a Boolean formula ϕ , we will demonstrate that determining a subset-minimal contrastive reason — whether for a neural network or a tree ensemble — allows us to decide the satisfiability of ϕ .

First, we check whether assigning all variables in ϕ to 1 evaluates ϕ to True. If so, the formula is satisfiable, and we have determined its satisfiability. Otherwise, we use Lemma 8 to encode the CNF formula as a neural network f . Next, we compute a subset-minimal δ -sufficient reason by setting $\mathcal{D}$ as the empirical distribution containing only a single data point $\mathbf{0}_n$, following a similar procedure to Lemma 6. Additionally, we set $\delta = 1$ and compute a subset-minimal δ -contrastive reason concerning $\langle f, \mathbf{x} \rangle$.

We will now demonstrate that if any subset-minimal δ contrastive reason obtained for ϕ is satisfiable, then ϕ itself is satisfiable. Conversely, if no subset-minimal δ contrastive reason is obtained, then ϕ is unsatisfiable. The validity of this claim follows a reasoning similar to that provided in Lemma 6. Specifically, the term $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$, where the distribution $\mathcal{D}$ considers sampling from a single datapoint, can set the probability to either 0 or 1. Furthermore, since we are searching for a $\delta = 1$ contrastive reason, this is equivalent to asking whether there exists an assignment that changes the classification of f , which corresponds to modifying the assignment of ϕ from False to True. If such an assignment exists, then ϕ is satisfiable. However, if no subset-minimal contrastive reason exists, then no subset of features fixed to zero — when the complementary set is set to ones — evaluates to true. This is equivalent to stating that no assignment evaluates f (and consequently ϕ) to True, implying that ϕ is unsatisfiable.

To extend the proof from neural networks to tree ensembles, we can follow the same procedure outlined in Lemma 6, encoding the CNF formula into an equivalent random forest classifier. Consequently, the proof remains valid for tree ensembles, thereby concluding the proof.

□

K PROOF OF THEOREM 3

Theorem 3. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, no bounded approximation exists for computing a local cardinally minimal δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. We divide the proof of the theorem into two lemmas, covering both the approximation guarantee and the result on the absence of a bounded approximation.

Lemma 10. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence.*

Proof. We will in fact prove a stronger claim, showing that this holds for *any* model, provided the trivial condition that its inference time is computable in polynomial time, along with one additional mild condition that we will detail later — both of which apply to both our neural network and tree ensemble formalizations.

We start by noting that, since we are working with empirical distributions, the computation of the global contrastive probability value function:

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (67)$$

Can be computed in polynomial time by iterating over all pairs $\mathbf{x}, \mathbf{z}$ in the dataset D , as previously established in Proposition 2. Since Algorithm 2 performs only a linear number of these polynomial-time queries, its total runtime is therefore polynomial.

Regarding the approximation, the classical work by Wolsey et al. (Wolsey, 1982) established a harmonic-series-based approximation guarantee for monotone non-decreasing submodular functions v with integer values. More generally, their result yields an approximation factor of $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(\{i\})}\right)\right)$. We showed in Proposition 2 that the value function v_{con}^g is monotone non-decreasing, and under feature independence, Proposition 4 establishes its submodularity. Combined with the fact that Algorithm 2 runs in polynomial time, this directly yields an approximation guarantee of $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$.

However, we must also show that the expression $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ is both finite and bounded by $\mathcal{O}(\ln(|D|))$, implying that it is effectively constant, due to the assumption of a fixed dataset $|D|$. To ensure this, we begin by confirming that the expression is well-defined and finite. We then proceed to establish the desired bounds. To do so, we introduce a preprocessing step in which we eliminate “redundant” elements—those that could theoretically cause the denominator $\min_{i \in [n]} v_{\text{con}}^g(\{i\})$ to be zero. We start by formally defining what we mean by redundancy:

Definition 1. *Let $\mathcal{D}$ denote some empirical distribution over a dataset D . Then we say that some feature $i \in [n]$ is redundant with respect to $\mathcal{D}$ if for any pair $\mathbf{x}, \mathbf{z} \in D$ it holds that $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) = f(\mathbf{x})$.*

Here, the notation $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}})$ indicates that all features in $[n] \setminus \{i\}$ are fixed to their values in $\mathbf{x}$, while feature i is set to its value in $\mathbf{z}$. Notably, this is equivalent to defining:

$$\forall S \subseteq [n], \mathbf{z} \in D \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}_{S \setminus \{i\}}; \mathbf{z}_{\bar{S} \cup \{i\}}) \quad (68)$$

As before, the notation $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ indicates that the features in S are fixed to their values in $\mathbf{x}$, while the features in $\bar{S}$ are fixed to their values in $\mathbf{z}$. Thus, in this sense, if we “remove” feature i from the input space and define a new function f' over the reduced space $[n'] := [n] \setminus \{i\}$, then for any input of size n' , the output of f' will exactly match the output of f when applied to the same features $[n] \setminus \{i\}$.

This removal can be done in polynomial time for both tree ensembles and neural networks: for neural networks, it involves detaching the corresponding input neuron from the network; for tree ensembles, it involves removing any splits on that feature from all decision trees. Given the empirical nature of the contrastive value function $v_{\text{con}}^g(S)$ — as previously discussed in this proof and in Proposition 2 — we can compute each $v_{\text{con}}^g(\{i\})$ for all $i \in [n]$ by iterating over all pairs $\mathbf{x}, \mathbf{z} \in \mathcal{D}$ and checking whether:

$$f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) \neq f(\mathbf{x}) \quad (69)$$

If this condition holds, it indicates that feature i is not redundant. Conversely, if the condition fails for all pairs considered during iteration, then i is deemed redundant. Once all redundant features have been identified with respect to the empirical distribution $\mathcal{D}$, we can remove them and construct an equivalent model f' . As discussed above, this transformation can be performed in polynomial time for both neural networks and decision trees.

To conclude our proof, we observe that $v_{\text{con}}^g(\{i\})$ equals 0 for some empirical distribution $\mathcal{D}$ if and only if feature i is redundant with respect to $\mathcal{D}$. Therefore, once the preprocessing step removes all redundant features from f and we construct the resulting function f' , we ensure that $\min_{i \in [n]} v_{\text{con}}^g(\{i\}) > 0$.

We next present the more precise bounds referenced in our proof. Specifically, we demonstrate that the approximation factor is $\mathcal{O}(\ln(|\mathcal{D}|))$. Given that the empirical distribution $\mathcal{D}$ is fixed, this yields a constant approximation. While tighter bounds may be achievable, our goal here is solely to establish that the bound is *constant* — a property that will later sharply contrast with the local explanation setting, where no bounded approximation exists.

More concretely, we know that the probability function $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$ is, by the definition of empirical distributions, at least $\frac{1}{|\mathcal{D}|}$. Therefore, the definition of $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$ (i.e., the value function v_{con}^g) is at least $\frac{1}{|\mathcal{D}|^2}$. In particular, this also implies that:

$$\min_{i \in [n]} v_{\text{con}}^g(\{i\}) \geq \frac{1}{|\mathcal{D}|^2} \quad (70)$$

Consequently, we obtain that:

$$\ln \left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})} \right) \leq \ln(|\mathcal{D}|^2) = \mathcal{O}(\ln(|\mathcal{D}|)) \quad (71)$$

Which concludes our proof.

□

Lemma 11. *Unless $\text{PTIME} = \text{NP}$, there is no bounded approximation for computing a cardinally minimal local δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$ where f is either a neural network or a tree ensemble, even when $|\mathcal{D}| = 1$.*

Proof. Assume, for contradiction, that there exists a bounded approximation algorithm for computing a cardinally minimal local δ -contrastive reason for some $\langle f, \mathbf{x} \rangle$, where f is a neural network or a tree ensemble—even when $|\mathcal{D}| = 1$. However, Lemma 9 establishes that even with a single baseline $\mathbf{z} = 0_n$ (i.e., when the entire dataset is just one instance), deciding whether a contrastive reason exists is NP-hard unless $\text{PTIME} = \text{NP}$. Therefore, if a polynomial-time algorithm could yield a bounded approximation for a cardinally minimal contrastive reason, it would contradict this result, as such an approximation would implicitly decide the existence of a contrastive explanation.

□

L PROOF OF THEOREM 4

Theorem 4. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O} \left(\frac{1}{1-k_f} + \ln \left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})} \right) \right)$ -approximation for*

computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence. In contrast, unless $PTIME=NP$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.

Proof. We divide the proof into two lemmas: one establishing the approximation guarantee for the global case, and the other demonstrating the intractability of the local case.

Lemma 12. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence.*

Proof. The proof will follow a similar approach to that of Lemma 10, where we established that for both neural networks and tree ensembles, Algorithm 2 achieves an approximation factor of $\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)$. After applying the preprocessing step, this ratio is guaranteed to be finite and bounded by $\mathcal{O}(\ln(|D|))$.

Similar to Lemma 10, we will again prove a stronger claim — namely, that the result holds for *any* model, assuming the trivial condition that its inference time is polynomially computable, and the additional condition concerning the removal of redundant features, as described in Lemma 10. As discussed there, both conditions are satisfied by our neural network and tree ensemble formalizations.

Here as well, since we are working with empirical distributions, the computation of the global sufficient probability value function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (72)$$

can be computed in polynomial time by iterating over all pairs $\mathbf{x}, \mathbf{z}$ in the dataset D , as previously shown in both Lemma 10 and Proposition 2. Since Algorithm 2 makes only a linear number of such polynomial-time queries, its overall runtime is polynomial.

Unlike Lemma 10, where the approximation relied on the result by Wolsey et al. (Wolsey, 1982) for monotone non-decreasing submodular functions, the setting here requires a different condition due to the supermodular nature of the function. Specifically, Shi et al. (Shi et al., 2021) provided an approximation guarantee of $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$, where $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$.

We established in Proposition 2 that the value function v_{suff}^g is monotone non-decreasing, and under the assumption of feature independence, Proposition 3 further shows that it is supermodular. Given that Algorithm 2 runs in polynomial time, this leads directly to an approximation guarantee of $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$, where $k^f := 1 - \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})}{v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset)}$.

To show that this expression is both bounded and constant, we follow the same preprocessing step as in Lemma 10, where we remove all redundant features from f — a process that, as previously explained, can be carried out in polynomial time for both neural networks and tree ensembles. As before, a feature i is strictly redundant if and only if $v(\{i\}) - v(\emptyset) = 0$. This preprocessing yields a new function f' that behaves identically to f over the remaining features and ensures that, for every i , both $v_{\text{suff}}^g(\{i\}) > 0$ and $v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset) > 0$ hold.

In order for us to show that this expression is both bounded and constant, similarly to Lemma 10 we will perform the exact same preprocessing phase (which, as we explained there, can be performed in polynomial time both for neural networks as well as tree ensembles) as before where we remove all redundant features from f . We note that here too, it holds that a feature i is strictly redundant if and only if $v(\{i\}) - v(\emptyset) = 0$. Hence, this preprocessing phase will give us a new function f' for which it both holds that it is equivalent to the behaviour of f for all remaining features and also satisfies that for any i it holds that both $v_{\text{suff}}^g(\{i\}) > 0$ and $v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset) > 0$ hold.

Now, following the same reasoning as in Lemma 10, where the probability term is lower bounded by $\frac{1}{|D|}$ and its expected value by $\frac{1}{|D|^2}$, we obtain that $\min_{i \in [n]} v_{\text{suff}}^g(i) \geq \frac{1}{|D|^2}$ and $v_{\text{suff}}^g(i) - v_{\text{suff}}^g(\emptyset) \geq \frac{1}{|D|^2}$. This implies, as in Lemma 10, that the first term is lower bounded by $\mathcal{O}(\ln(|D|))$, and for the second term we derive the following:

Since v_{suff}^g is supermodular (Proposition 3), we have that $v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})$. Therefore $1 \leq v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\}) \leq \frac{1}{|D|^2}$. This implies: $1 \leq \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})}{v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset)} \leq |D|^2$, and therefore, $1 - |D|^2 \leq k^f \leq 0$. This yields $\frac{1}{1 - k^f} \leq |D|^2$, demonstrating that the overall approximation bound based on the one established by Shi et al. is constant. While this bound may be significantly smaller in practice, our goal here is simply to show that it remains constant — unlike for the local formulation, where no bounded approximation is achievable.

□

Lemma 13. *Unless $\text{PTIME} = \text{NP}$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. The proof proceeds a similar approach to Lemma 11. Suppose, for contradiction, that there exists a polynomial-time algorithm that provides a bounded approximation to a cardinally minimal local δ -sufficient reason for some $\langle f, \mathbf{x} \rangle$, where f is either a neural network or a tree ensemble—even when $|D| = 1$. Yet, Lemma 6 shows that even in the extreme case where the dataset consists of a single baseline $\mathbf{z} = 0_n$, deciding whether a sufficient reason exists is NP-hard unless $\text{PTIME} = \text{NP}$. Hence, the existence of such an approximation algorithm would contradict this hardness result, as it would entail the ability to decide the existence of a sufficient reason.

□

M HARDNESS OF APPROXIMATING GLOBAL CONTRASTIVE REASONS FOR NEURAL NETWORKS USING EMPIRICAL DISTRIBUTIONS

Proposition 1. *Assume a neural network $f : \mathbb{F} \rightarrow [c]$ with ReLU activations, some dataset $\mathbf{D}$, such that $\mathcal{D}$ is the empirical distribution defined over $\mathbf{D}$, and some threshold $\delta \in \mathbb{Q}$. Then unless $\text{NP} \subseteq \text{DTIME}(n^{\mathcal{O}(\log(\log(n)))})$, there does not exist a polynomial-time $\mathcal{O}(\ln(|\mathbf{D}|) - \epsilon)$ approximation algorithm for computing a cardinally minimal global contrastive reason for f concerning $\mathcal{D}$, for any fixed $\epsilon > 0$.*

Proof. The proof proceeds via a polynomial-time approximation-preserving reduction from a specific variant of Set Cover that is, similarly to regular Set Cover, provably hard to approximate within the factor given in the proposition (Vazirani, 2001).

Partial Set Cover:

Input: A universe $U := \{u_1, \dots, u_n\}$, a collection of sets $S := \{S_1, \dots, S_m\}$ such that for all j $S_j \subseteq U$, and a coverage threshold $\delta \in \{0, 1, \dots, n\}$.

Output: A minimum-size subset $C \subseteq S$ such that $|\cup_{S_j \in C} S_j| \geq \delta$.

We reduce this problem to that of computing a *global*, cardinally-minimal contrastive reason S for a neural network f with respect to a distribution $\mathcal{D}$ defined over a dataset $\mathbf{D}$, and some threshold δ' . We start by recalling the definition of a global contrastive reason and then extend it to the setting where the explanation is evaluated over an empirical dataset $\mathbf{D}$. The definition is as follows:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}} [\Pr_{\mathbf{z} \sim \mathcal{D}} (f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (73)$$

More concretely, when we assume an empirical distribution over $\mathbf{D}$, this implies that:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} [\Pr_{\mathbf{z} \sim \mathcal{D}} (f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] = \\ & \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} [\Pr_{\mathbf{z} \sim \mathcal{D}} (f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] = \\ & \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} \left(\frac{\sum_{\mathbf{z} \in \mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \right) \end{aligned} \quad (74)$$

where we denote $\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x}) := \{\mathbf{z} \in \mathbf{D} : \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}\}$. We next construct a ReLU neural network f . Before doing so, we specify the functional form that we intend the constructed neural network f to

implement. For each set S_j , we introduce two vectors $\mathbf{x}^j$ and $\mathbf{y}^j$, both belonging to $\mathbb{Z}^n$ and drawn from the dataset $\mathbf{D}$, such that they agree on all coordinates outside S_j ; that is, they share identical feature values on the complement $\bar{S}_j$. Formally:

$$\mathbf{x}_{\bar{S}_j}^j = \mathbf{y}_{\bar{S}_j}^j \quad (75)$$

and we assign strictly different values to the features in S_j by selecting them from two distinct vectors $\mathbf{z}$ and $\mathbf{z}'$ such that:

$$\mathbf{x}^j = (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = (\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = \mathbf{y}^j \quad (76)$$

We aim to construct f such that it satisfies the following:

$$f(\mathbf{x}^j) = f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = f(\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = f(\mathbf{y}^j) \quad (77)$$

We next assume that both $\mathbf{x}^j$ and $\mathbf{y}^j$ occur *uniquely* in the dataset $\mathbf{D}$ across all $[n]$ coordinates — that is, no other vector in $\mathbf{D}$ matches either of them on any coordinate. Under this assumption, we obtain the following relation:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{S_j} = \mathbf{x}_{S_j})] = \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} \left(\frac{\sum_{\mathbf{z} \in \mathbf{D}_{S_j}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathbf{D}_{S_j}(\mathbf{z}, \mathbf{x})} \right) = \frac{2}{2|\mathbf{D}|} = \frac{1}{|\mathbf{D}|} \quad (78)$$

Here, the factor of 2 appearing in both the numerator and denominator arises from the fact that only two vector pairs — $(\mathbf{x}^j, \mathbf{z}^j)$ and $(\mathbf{z}^j, \mathbf{x}^j)$ — contribute to the respective sums.

We repeat this construction for each S_i in the set-cover instance, each time choosing two *distinct* vectors $\mathbf{x}^i$ and $\mathbf{y}^i$ that differ on *every* coordinate in $[n]$. In total, this yields $2m$ vectors that form the dataset $\mathbf{D}$ from which we construct f .

We then construct a neural network f that simulates this function. This is achieved using a known result on *neural network memorization* (Vardi et al., 2022), which guarantees that a ReLU network capable of representing any finite set of input–output pairs can be built in polynomial time. Using this result, we construct f accordingly, and set $\delta' := \frac{\delta}{|\mathbf{D}|}$.

We next argue that a subset S_j is a global contrastive reason for f under tolerance δ' if and only if C is a partial set cover for U under tolerance δ . By Equation 78, each constructed pair of vectors $(\mathbf{x}^j, \mathbf{y}^j)$ contributes exactly $\frac{1}{|\mathbf{D}|}$ to the value term. Moreover, because each S_j uses its own distinct pair of vectors in $\mathbb{Z}^n$, there is no overlap between these pairs. Consequently, the coverage achieved by any set cover C corresponds exactly to the accumulated contribution of the selected subsets in the input domain of f . This completes the reduction.

□

N TRACTABILITY RESULTS FOR ORTHOGONAL DNFS

Lemma 14. *If f is an orthogonal DNFs and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time.*

Proof. We start by defining orthogonal DNFs (Crama & Hammer, 2011) before presenting the full proof.

Definition 2 (Orthogonal DNFs). *A Boolean function φ is in orthogonal disjunctive normal form (orthogonal DNF) (Crama & Hammer, 2011) if it is a disjunction of terms $T_1 \vee T_2 \vee \dots \vee T_m$, such that for every pair of distinct terms T_i and T_j ($i \neq j$):*

$$T_i \wedge T_j \models \perp$$

Equivalently, no single variable assignment can satisfy more than one term simultaneously.

2214 The proof follows the same general argument as Theorem 1, relying on the fact that v_{suff}^g can be
 2215 computed in polynomial time for any set S .

2216 For a given set S , let $\mathcal{C}^S T_i$ be the constraints that the term T_i imposes on the variables in S .
 2217 We enumerate all pairs of terms (T_i, T_j) . Any pair whose constraints on S are *inconsistent* (i.e.,
 2218 $\mathcal{C}_{T_i}^S \wedge \mathcal{C}_{T_j}^S \models \perp$) is ignored, since it contributes zero to v_{suff}^g .
 2219

2220 For each *consistent* pair of terms (T_i, T_j) , we evaluate $\Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S)$, where $\mathcal{C}_{T_i, T_j}^S = \mathcal{C}_{T_i}^S \cap \mathcal{C}_{T_j}^S$.
 2221 The contribution of this pair to v_{suff}^g is then
 2222

$$2223 \Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_i}^{\bar{S}}) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_j}^{\bar{S}}).$$

2224
 2225 Under feature independence, these probabilities factor as $\Pr(\mathbf{x}_S) = \prod_{i \in S} p(x_i)$. Since there are at
 2226 most m^2 such term pairs, the value of v_{suff}^g can be computed in polynomial time under an independent
 2227 distribution $\mathcal{D}$. $\square$
 2228

2229 O DISCLOSURE: USAGE OF LLMs

2230
 2231 An LLM was used exclusively as a writing assistant to refine grammar, correct typos, and improve
 2232 clarity and framing. It did not contribute to research ideas, study design, data analysis, or interpretation
 2233 of results, all of which were carried out solely by the authors.
 2234

2235
 2236
 2237
 2238
 2239
 2240
 2241
 2242
 2243
 2244
 2245
 2246
 2247
 2248
 2249
 2250
 2251
 2252
 2253
 2254
 2255
 2256
 2257
 2258
 2259
 2260
 2261
 2262
 2263
 2264
 2265
 2266
 2267